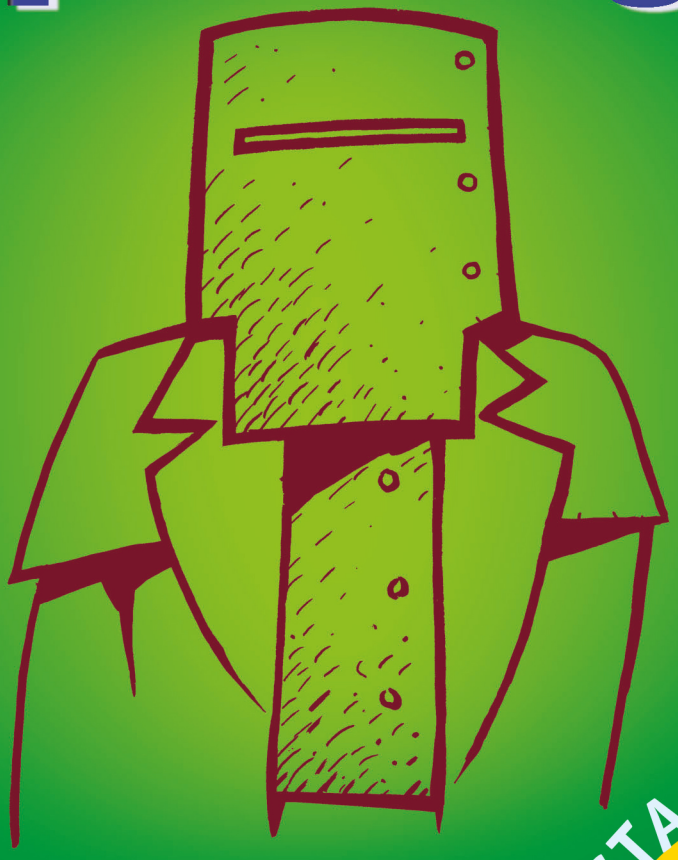
MATHS in Crime & Espionage



Denis Day

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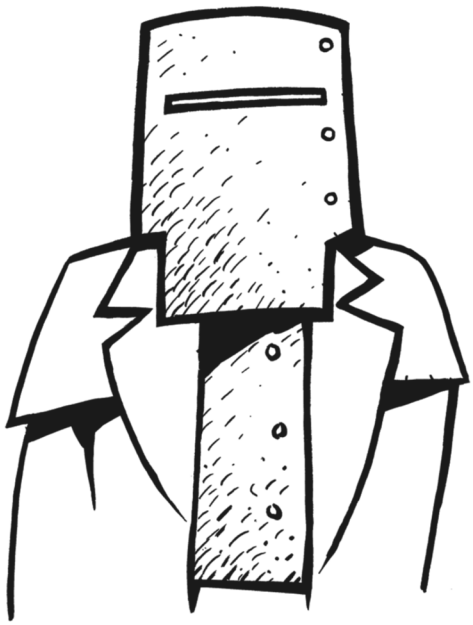
MATHS in Crime & Espionage



Denis Day

This book was commissioned by Jennifer Gough, project management, editing and typesetting was by Alison Stone, illustrations by Eddy Crosby, and Gail McManus Graphics designed the cover.

Denis Day completed a mechanical engineering diploma and worked as a production engineer before completing his education qualifications (Master of Education). For more than 30 years he has been teaching maths and science in socially disadvantaged schools and he has also taught in private schools. Denis has presented numerous workshops at the Maths Association of Victoria's annual maths conference and regional conferences, and developed many curriculum materials.

His particular strength is in devising high interest activities which immediately provide a meaningful context for the mathematical concepts being studied – making learning both more effective and much more enjoyable for students.

Denis's interests include playing and coaching tennis (Level 2 tennis coach), saltwater fishing and Aussie Rules football (Level 2 umpire).

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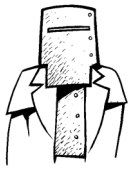
Introduction



Maths in Crime & Espionage is a collection of photocopiable worksheets that apply mathematical knowledge and techniques to exciting investigations of real life and fictional crimes. Students learn about heroic outlaws, spies and forensics. Each topic has been written to interest and engage students in Years 9–10, and features content that requires students to apply key maths skills such as basic operations, measurement, graphs, percentage, use of formula and analysis of data. The activities will help students to see how the skills of maths can be used to understand real world events.

Many of the activities in this book utilise data from a variety of sources. This data may have been reconfigured or altered for the purpose of understanding an essential mathematical concept. Whilst approximately correct, this data is not intended to be accurate and students should not use any data or statistics as source material for other projects.

Unless advised otherwise, students are asked to round their answers to two decimal places. For some topics a calculator would be useful. Answers are provided to make marking quick and easy.

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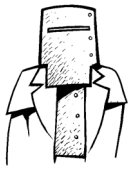
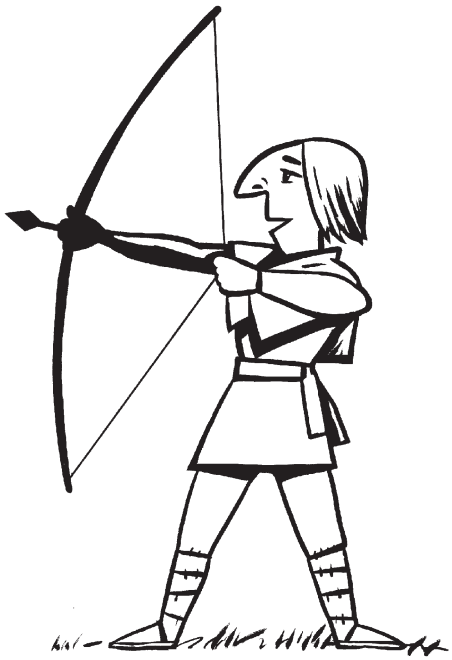
Robin Hood



1

Robin Hood is a heroic outlaw who is said to have lived in England in the thirteenth century. He and his gang of merry men are renowned for stealing from the rich to give to the poor. His merry men included Will Scarlet, Much the Miller's Son and Little John. Historians have not been able to prove whether Robin Hood was a real man or a man created as a fictional story for entertainment. The tales of Robin Hood have certainly entertained people for many years.

From 1228 onwards the names 'Robinhood', 'Robehod' or 'Hobbehod' occur in the rolls of several English Justices. Between 1261 and 1300 there are at least eight references to 'Rabunhod' in various regions across England.



1 It is the year 2012. How many years have passed since some variation of the name Robin Hood began to occur?

2 Over how many years did the eight references to 'Rabunhod' appear?

There are many discrepancies about the birth and life of Robin Hood in the various tales. One manuscript says that he was born in 1160. Another historian dates his birth at 1225, and yet another manuscript states that he was born between 1285 and 1295.

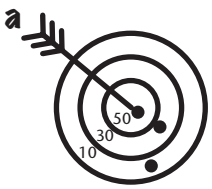
3 What is the range in the years of the estimates of Robin Hood's birth date?

4 Why do you think there is a large variation in the possible birth dates of Robin Hood?

Robin Hood used a longbow as his weapon. It was important to be skilled at the bow and arrow in the 13th and 14th century because it was a means of hunting, survival and protection.

5 Calculate Robin Hood's score on each set of targets.

a



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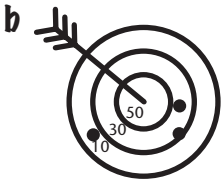

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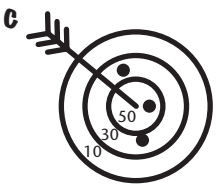

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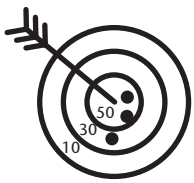
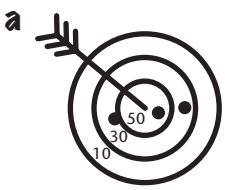
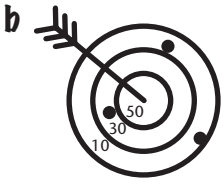


6 Calculate Little John's score on each set of targets.

a



b



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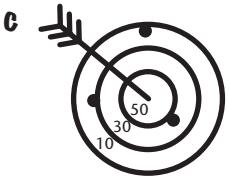

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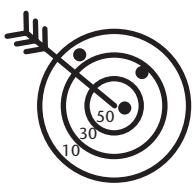

7 Who was the better marksman? Why?

The City of Nottingham, the setting of many of the Robin Hood legends, was originally named after an Anglo-Saxon Chieftain. In 1068 the Normans built Nottingham Castle.

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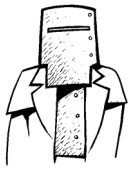
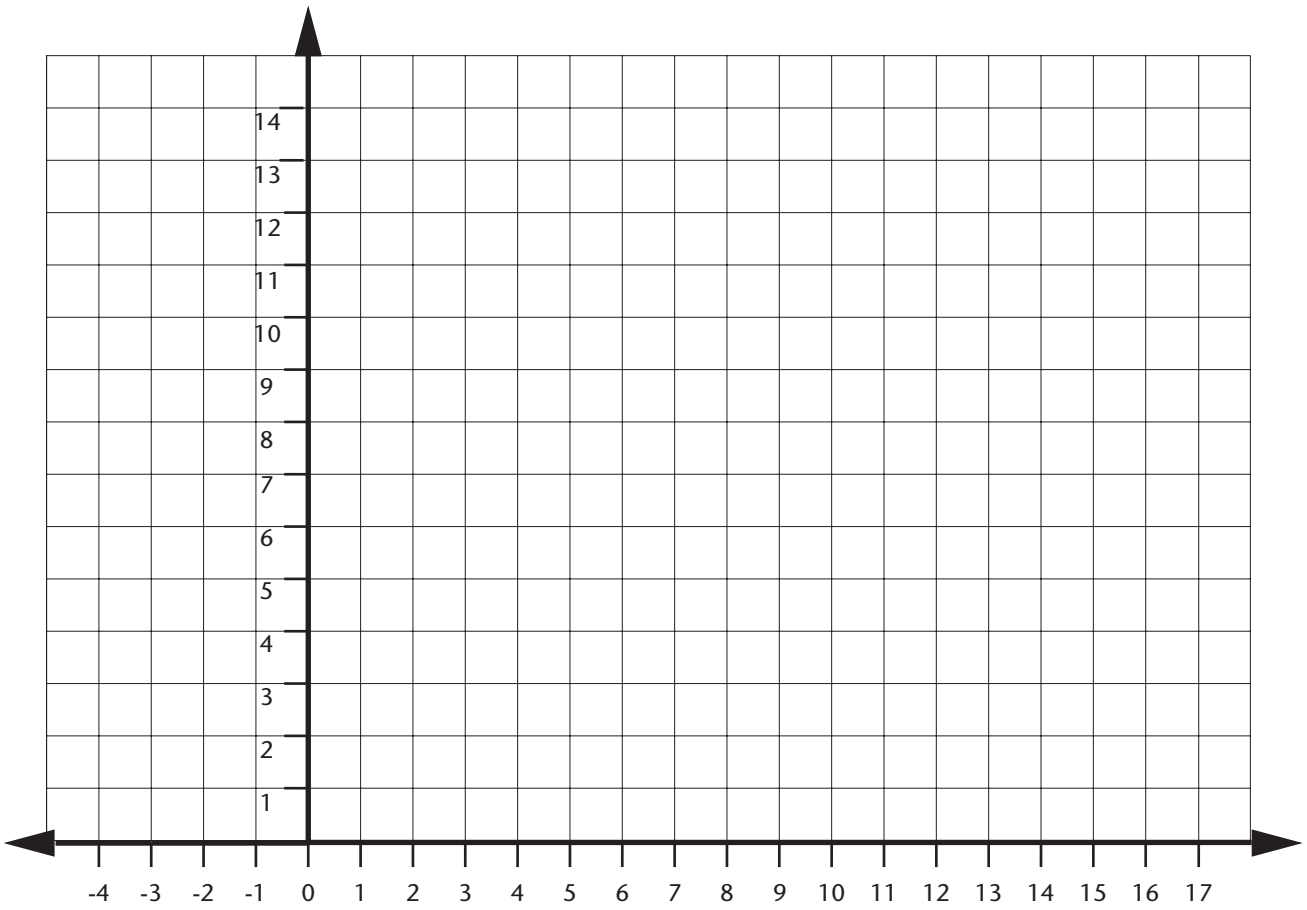


8 Connect the points to form a picture.

Start: (5, 12) (5, 14.5) (7, 14) (5, 13.5) stop. Start: (4, 0) (4, 8) (5, 9) (5, 10) (5.5, 10.5) (5.5, 12) (4, 12) (4, 11.5) (3.5, 11.5) (3.5, 12) (2.5, 12) (2.5, 11.5) (2, 11.5) (2, 12) (1, 12) (1, 11.5) (0.5, 11.5) (0.5, 12) (-0.5, 12) (-0.5, 10.5) (0, 10) (0, 3.75) stop. Start: (-4, 0) (-4, 3.5) (-3.5, 3.5) (-3.5, 3) (-3, 3) (-3, 3.5) (-2, 3.5) (-2, 3) (-1.5, 3) (-1.5, 3.5) (-0.5, 3.5) (-0.5, 3) (0, 3) (0, 3.5) (1, 3.5) (1, 3) (1.5, 3) (1.5, 3.5) (2.5, 3.5) (2.5, 3) (3, 3) (3, 3.5) (4, 3.5) stop. Start: (4, 8) (5, 8) (5, 7.5) (5.5, 7.5) (5.5, 8) (6.5, 8) (6.5, 7.5) (7.5, 7.5) (7.5, 8) (8, 8) (8, 0) stop. Start: (16, 6) (16.5, 6) (16.5, 7) (16.25, 7.5) (16, 7) (16, 6) stop. Start: (-4, 2) (4, 2) stop. Start: (15, 3) (15, 4.5) (15.5, 5) (16, 4.5) (16, 3) (15, 3) stop. Start: (12, 9) (13.5, 9) (13.5, 7.5) (12.5, 6.5) (12.5, 2) stop. Start: (14, 6) (14, 7) (14.25, 7.5) (14.5, 7) (14.5, 6) (14, 6) stop. Start: (13.5, 8) (17, 8) (17, 17.5) (13.5, 7.75) stop. Start: (8, 8) (7, 9) (5, 9) stop. Start: (13, 5.5) (13, 2.5) stop. Start: (12, 5.75) (17, 5.75) (17, 5.5) (12, 5.5) stop. Start: (0.5, 7.5) (0.5, 9) (1, 9.5) (1.5, 9) (1.5, 7.5) (0.5, 7.5) stop. Start: (3, 9.5) (3.5, 9) (3.5, 7.5) (2.5, 7.5) (2.5, 9) (3, 9.5) stop. Start: (7.5, 9) (7.5, 10) (8, 10.5) (8.5, 10) (8.5, 9) (7.5, 9) stop. Start: (7, 9) (7, 11) (6.5, 11.5) (6.5, 13) (7.5, 12.5) (8, 12.5) (8, 13) (9, 13) (9, 12.5) (9.5, 12.5) (9.5, 13) (10.5, 12.5) (9.5, 13) (10.5, 12.5) (11, 12.5) (11, 13) (12.5, 13) (12.5, 11.5) (12, 11) (12, 0) stop. Start: (6.5, 11.5) (12.5, 11.5) stop. Start: (-0.5, 10.5) (5.5, 10.5) (5, 10) (0, 10) stop. Start: (4, 4.25) (12, 4.25) (12, 4) (4, 4) stop. Start: (15, 6) (15, 7) (15.25, 7.5) (15.5, 7) (15.5, 6) (15, 6) stop. Start: (13.5, 3) (13.5, 4.5) (14, 5) (14.5, 4.5) (14.5, 3) (13.5, 3) stop. Start: (10, 8.5) (11, 7.5) (11, 5) (9, 5) (9, 7.5) (10, 8.5) stop. Start: (10.5, 9) (11.5, 9) (11.5, 10) (11, 10.5) (10.5, 10) (10.5, 9) stop. Start: (3, 6) (3, 4) (1, 4) (1, 6) (2, 7) stop. Start: (12, 13) (12, 14.5) (15, 14) (12, 13.5) stop. Start: (5, 0) (5, 2.5) (6, 3.5) (7, 2.5) (7, 0) stop.

Add a large bush at (14.5, 0). Colour the picture.



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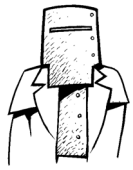
Ned Kelly



2

Time has clouded the legend of Edward 'Ned' Kelly, an Irish-Australian bushranger of the late 19th century. Ned, along with a band of accomplices known as The Kelly Gang, became symbolic for his daring defiance of authority. The debate about Ned's status as ruthless criminal or victimised folk hero will most likely continue throughout history.

Edward (Ned) Kelly was born in Beveridge, just north of Melbourne, between June 1854 and June 1855. In 1869, he was arrested for assaulting a Chinese pig farmer. A year later Ned was arrested for assaulting Jeremiah McCormack and late that same year was sentenced to three years' imprisonment for receiving a stolen horse. In September 1877 Ned was arrested for drunkenness and in October of the following year the infamous exchange at Stringybark Creek saw Ned shooting Sergeant Kennedy dead. Later that same year the Kelly Gang robbed the National Bank at Euroa of £2260. The following year the Kelly Gang robbed the local bank in Jerilderie of £2414. In June 1880 the Kelly Gang took about 70 hostages at the Glenrowan Inn, before being captured. In November of that same year Ned Kelly was hanged at the Old Melbourne Gaol for the murder of Constable Lonigan.

1 Draw a timeline that shows these events in the life of Ned Kelly.

2 If Ned was born in June 1854, how old was he, in years and months, when he was sentenced to 3 years of hard labour?

3 How old was Ned when he was hanged, if he was born in June 1854?

4 How much time did it take between the capture of Ned and his hanging at the Old Melbourne Gaol?

5 There seems to be some debate as to the actual birth date of Ned Kelly. John Kelly made the notes shown in the table when he registered the birth of Ned's sister Grace, on the 3rd October 1865. Complete the table.

	Name	Date of birth	Age (years & months)
a	Anne	November 1853	
b	Edward (Ned)		10
c	Margaret	15/06/1857	
d	James	31/07/1859	
e	Daniel	01/05/1861	
f	Catherine	12/07/1863	
g	Grace	10/08/1865	

6 When Ned's father died on the 27th December 1866, Ned compiled the information in the table at the top over the page. Complete the table.

7 How accurate were Ned's estimates?

8 Considering your answers to Q5 and Q6, when do you think Ned was born? Why?

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	Name	Actual age	Ned's estimate of age
a	Anne		13
b	Edward (Ned)		11 years 6 months
c	Margaret		9 years
d	James		7 years
e	Daniel		5 years
f	Catherine		4 years
g	Grace		1 year 6 months

In 1879, the New South Wales government offered a reward of £4000 and the Victorian government offered £4000 for the capture of the Kelly Gang.

9 How much is the reward in total?

10 How much is the total reward in dollars if £1 = $2?

11 Assuming the average inflation rate over the years is 3.2%, find the value in 2012 of the reward offered for the capture of the Kelly Gang. (Use the rule $A = P\ (1 + R)^n$, where A = 2012 value, P = reward, R = rate of inflation as a decimal and n is the number of years).

12 The Kelly gang stole £2260 in 1878, and £2414 in 1879, from various banks. Convert these amounts to:

a dollars **b** dollars in 2012 value.

In 1880, there was considerable controversy over the division of the £8000 of reward money.

13 Complete the table below.

	Person	Amount in £	Total amount in £	Total amount in $	Total amount in $ (2012)
a	6 indigenous trackers	50 each			
b	5 assistant train drivers			620	
c	2 train engineers	95 each			
d	4 train operators		172		
e	9 senior police officers	276 each			
f	3 civilians	130 each			
g	2 senior constables	175 each			
h	1 detective			214	
i	17 constables	158 each			
j	1 sub-inspector	306			
k	1 superintendent	705			

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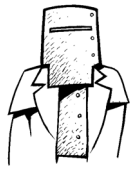
Ned Kelly rides again



3

The members of the Kelly gang were all equipped with armour that was tough enough to block bullets. Each man's armour weighed about 45 kilograms, and each one had a helmet.

1 Forty-five kilograms is very heavy, it is a little more than twenty, two-litre cartons of milk. Write down two other weights which are similar in weight to 45 kg.

2 Convert these weights to grams.

a 45 kg **b** 40 kg **c** 45.2 kg **d** 46.3 kg

3 Convert these weights to kilograms.

a 3000 g **b** 48 000 g **c** 4560 g **d** 56 500 g

The map shows the localities frequented by the Kelly Gang.

4 Approximately how far by road is it from Melbourne to:

a Beveridge? **b** Jerilderie?

c Beechworth? **d** Glenrowan?

5 The average speed a horse can travel over long distances is approximately 6 km/h. How long would it take Ned to travel between:

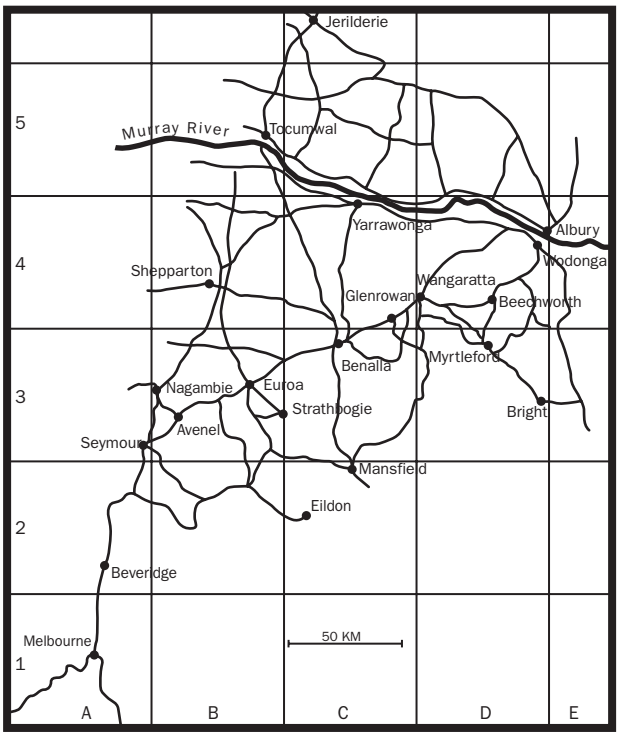
a Benalla and Jerilderie?

b Glenrowan and Beechworth?

c Avenel and Euroa?

d Shepparton and Nagambie?

e State one assumption you had to make in these calculations.



6 Ned was involved in major crime at Jerilderie, Glenrowan and Euroa. Locate and colour these towns on the map.

7 Find the town located at:

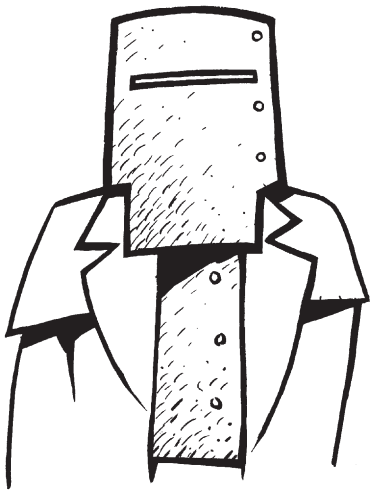
a B4 **b** C6 **c** B5

d D3 **e** A1.

8 Give the grid reference for:

a Beveridge **b** Wodonga

c Mansfield **d** Glenrowan.



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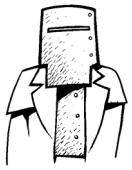
Ned Kelly rides again (cont'd)



3

9 Find each of these maths words in the grid below. The words are horizontal, vertical and diagonal and some are reversed.

CALCULATE DISTANCE INFLATION SPEED NUMBER TOTAL
VALUE GRAM DECIMAL KILOGRAM AGE RULE
AVERAGE DOLLAR POUND FREQUENT KILOMETREPERHOUR
LITRE LOCATION LENGTH FAR MILLILITRE TIMELINE

D	E	C	I	M	A	L	E	Q	A	R	E	G	D	I	R	E	V	E	B
M	I	L	L	I	L	I	T	R	E	E	G	H	I	L	X	A	S	G	Q
E	I	R	E	D	L	I	R	E	J	W	D	G	U	J	K	L	P	L	T
G	Q	D	R	T	J	I	N	F	L	A	T	I	O	N	D	E	E	K	Y
J	H	E	L	M	E	T	Q	W	E	R	E	J	N	U	M	B	E	R	H
T	D	R	E	T	H	F	V	H	E	D	F	A	H	J	R	W	D	G	S
M	P	O	U	N	D	D	B	T	H	J	L	Q	V	A	V	X	Z	T	H
E	R	G	R	T	U	J	L	I	T	R	E	T	T	E	K	L	O	H	T
K	I	L	O	M	E	T	R	E	P	E	R	H	O	U	R	W	R	O	G
R	E	E	A	W	T	E	H	W	R	T	I	O	T	P	M	A	G	R	N
E	R	N	D	E	H	W	F	C	O	N	S	T	A	B	L	E	G	S	E
G	T	R	I	Y	U	I	T	E	A	S	A	J	L	T	U	K	J	E	L
I	U	O	S	B	J	G	T	W	G	F	R	E	Q	U	E	N	T	E	K
S	I	W	T	V	V	A	L	U	E	A	M	C	N	H	Y	I	H	F	I
T	N	A	A	R	L	O	R	V	L	R	N	I	G	A	N	Y	Y	S	L
E	M	N	N	U	K	L	H	A	D	O	L	L	A	R	Y	H	H	F	O
R	G	E	C	S	L	G	U	Q	T	T	I	O	O	Y	A	G	R	O	G
T	T	L	E	X	M	A	R	U	L	E	U	P	W	A	R	M	O	U	R
M	A	B	Q	L	O	C	A	T	I	O	N	G	E	E	R	N	E	P	A
C	D	N	S	E	T	I	M	E	L	I	N	E	T	F	H	B	S	M	M

10 Find some other Ned Kelly words hidden in the grid above.

11 The answers to these questions are in the list of words that can be found in the above grid.

a What is the name of a unit that measures the speed of a car?

b What is it called when the price of groceries continually rises over time?

c A simple graphic drawn to show the historical events occurring over time is a _____ .

d What is another name for one thousand grams?

e What is another word similar to distance?

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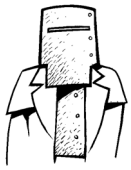
The Old Melbourne Gaol



4

Construction of the Old Melbourne Gaol began in 1839 and many extensions were added over the years to cope with the increasing numbers of prisoners. When it opened in 1845, it held fifty-nine men and nine women.

1 The contractor for the initial building received approximately £20 000 in 1854. If £1 = $2, how much did the contractor receive to build the gaol in dollars? Convert this to find the dollar value in 2012.

Use the rule $A = P\ (1 + R)^n$, where A = 2012 value, P = reward, R = rate of inflation as a decimal and n is the number of years. Assume the rate of inflation has averaged 3.2% during this time.

2 Various extensions were made to the Gaol between 1857 and 1864. Approximate costs involved with some of these construction works are shown in the table below. Calculate the cost of each stage and the total cost of construction in 2012 dollar values. Complete the table.

	Works	Amount in £	Amount in $ (1859)	Amount in $ (2012)
a	Two wings	10 000		
b	Other construction	18 000		
c	Boundary walls	25 000		
d	Staff quarters	9 000		
e	Watch tower	1 300		
f	North wing	16 000		
g	Extensions	17 000		
h	Total			

3 Calculate the total amount in 2012 dollar value that it would cost to build the Gaol. Don't forget to include the contractor's costs (paid in 1854). Write your answer in millions of dollars.

4 Some of the staff and approximate wages of the Gaol in 1847 were:

	Staff	Wage in £	Wage pa in $ (1847)	Wage pa in $ (2012)
a	Gaoler	135 pa		
b	Clerk	70 pa		
c	Doctor	90 pa		
d	Chaplain	75 pa		
e	Magistrate	50 pa		
f	Superintendent	30 pa		
g	Other staff (each)	20 pa		

Note: pa means per annum or per year.

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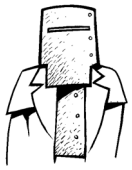
The Old Melbourne Gaol (cont'd)



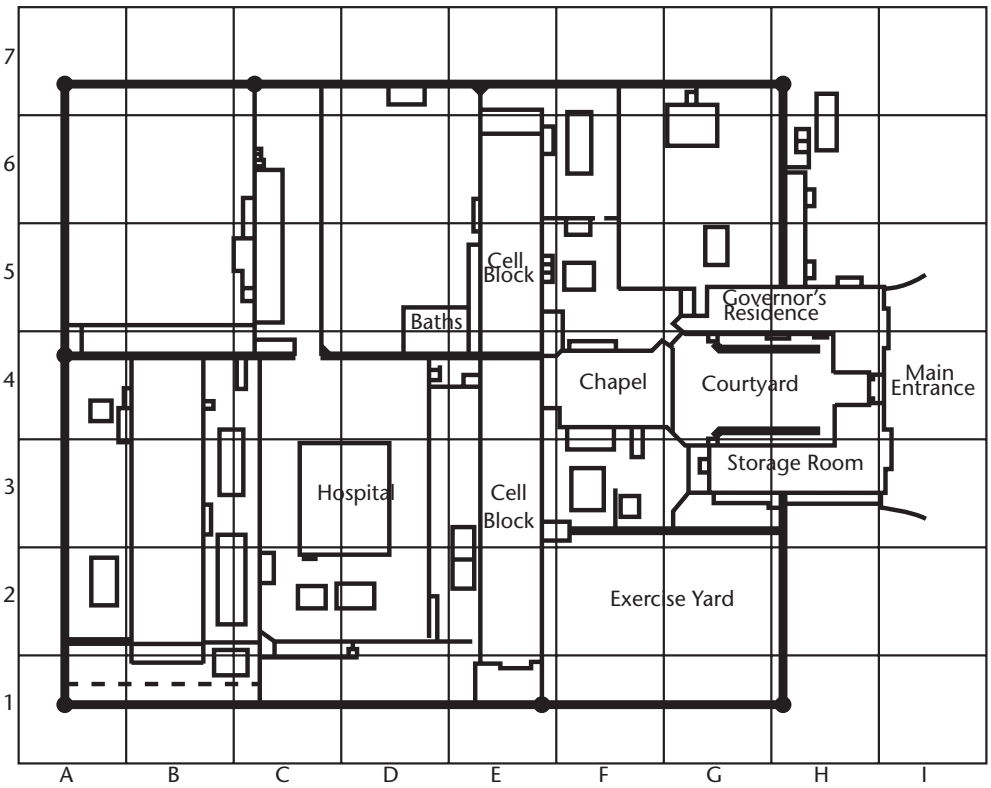
5 Do you think wages have grown faster than 3.2% pa since 1847? Explain your answer.

6 Find the total annual wage for the Gaol in today's dollar value.

7 Which workers earned the most in a year?

8 Which two workers earned the least in a year?

The diagram is the layout of the Old Melbourne Gaol.

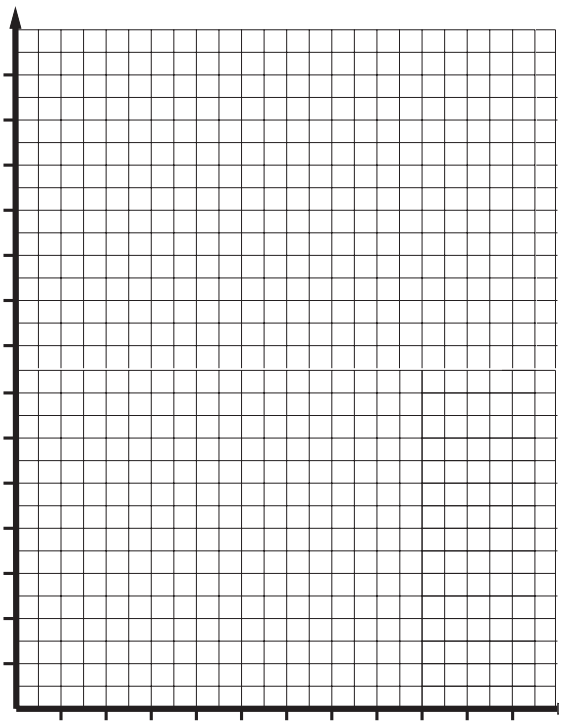


9 Name the feature located at the grid reference:

- **a** F2
- **b** C3
- **c** E3
- **d** H5
- **e** I4.

10 The table shows the number of hangings recorded at the Gaol and the crime of each prisoner hung. Draw a column graph to display the data.

Crime	Number hanged
Murder	63
Shooting	3
Arson	1
Robbery	2
other	21



11 How many recorded hangings are listed?

12 Historians claim there were 135 hangings at the Gaol. What percentage of the hangings has been recorded?

13 From the recorded data, what percentage of criminals were hanged for murder?

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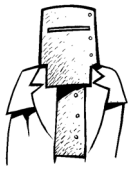
Jack the Ripper



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Jack the Ripper is the most notorious serial killer in history. He is believed to be responsible for a number of murders in the slum areas around the Whitechapel district of London in 1888. The name 'Jack the Ripper' originated in a letter, which was printed in the newspapers, written by someone claiming to be the murderer. Despite lengthy police investigations, the identity of the person who committed these crimes has never been discovered.

Date	Time of discovery	Victim	Place
Friday 31 August	3.40 am	Mary Ann Nichols, 44	Bucks' Row
Saturday 8 September	6.00 am	Annie Chapman, 47	29 Hanbury Street
Sunday 30 September	1.00 am	Elizabeth Stride, 45	Berner Street
Sunday 30 September	1.45 am	Catherine Eddowes, 46	Mitre Square
Friday 9 November	10.45 am	Mary Jane Kelly, 25	13 Miller's Court

1 Draw a timeline to show when the murders took place.

2 How many weeks separated each of the murders?

3 Calculate the average age of the victims.

The police interviewed more than 2000 people and of these, investigated almost 300 people and detained 80 of these people.

4 What is the fraction of people:

a investigated, who were interviewed?

b detained, who were investigated?

5 Calculate the percentage of people:

a investigated, who were interviewed

b detained, who were investigated.

6 How much time, in hours, did the police spend interviewing people if each interview took:

a 45 min? **b** 35 min?

c 2 h 35 min? **d** 3.2 h?

7 How much time, in hours, did the police spend investigating suspects if each investigation took:

a 5 days? **b** 4 days?

c 6 days? **d** 7.3 days?

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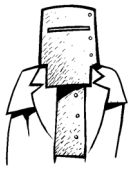
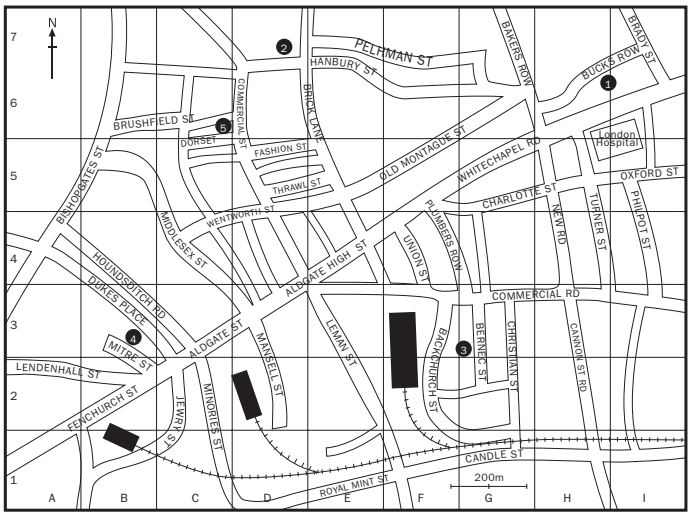
Jack the Ripper (cont'd)



This is a street map of the Whitechapel district where the Ripper crimes were committed. Use the map to answer the questions.



8 Name the streets located at

a F5 **b** G1.

9 Write down the name of each victim and the grid reference of the locality the body was found.

10 What feature is located in I5?

11 Estimate the distance between each victim that Jack the Ripper would have to walk by road. Write your answer in kilometres.

12 Draw a rectangle to enclose the victims. Calculate the area in which the victims were found.

13 What direction are you travelling when moving from:

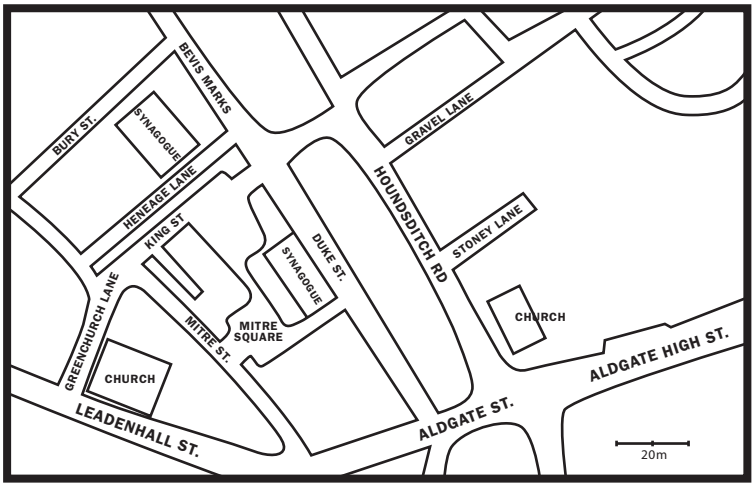
a Aldgate St along Whitechapel Rd? **b** Aldgate St along Commercial Rd?

c Bishop Gate Rd along Houndsditch Rd? **d** Candle St along Leman St?

14 Write down a set of directions to travel from London Hospital to victim 5.

Below is a more detailed map of the area where the body of Catherine Eddowes was found. Constable Watkin's beat was in this area.

15 Use arrows to draw Watkin's beat on the map. He started at the corner of Heneage Lane and Bevis Marks. He walked down Heneage Lane and turned left into Creechurch Lane, then left into Leadenhall Street. He then turned left into Mitre Street, around Mitre Square, before continuing along Mitre Street. He then turned right into King Street and then left into Bevis Marks.



16 Estimate the distance that Constable Watkins had to walk to complete his rounds.

17 If the average walking speed for a person is 5 km/h, how long would it take Watkins to complete one of his rounds?

18 How many rounds would he complete in an 8-hour shift, if he had no interruptions?

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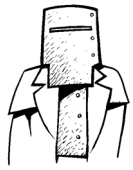
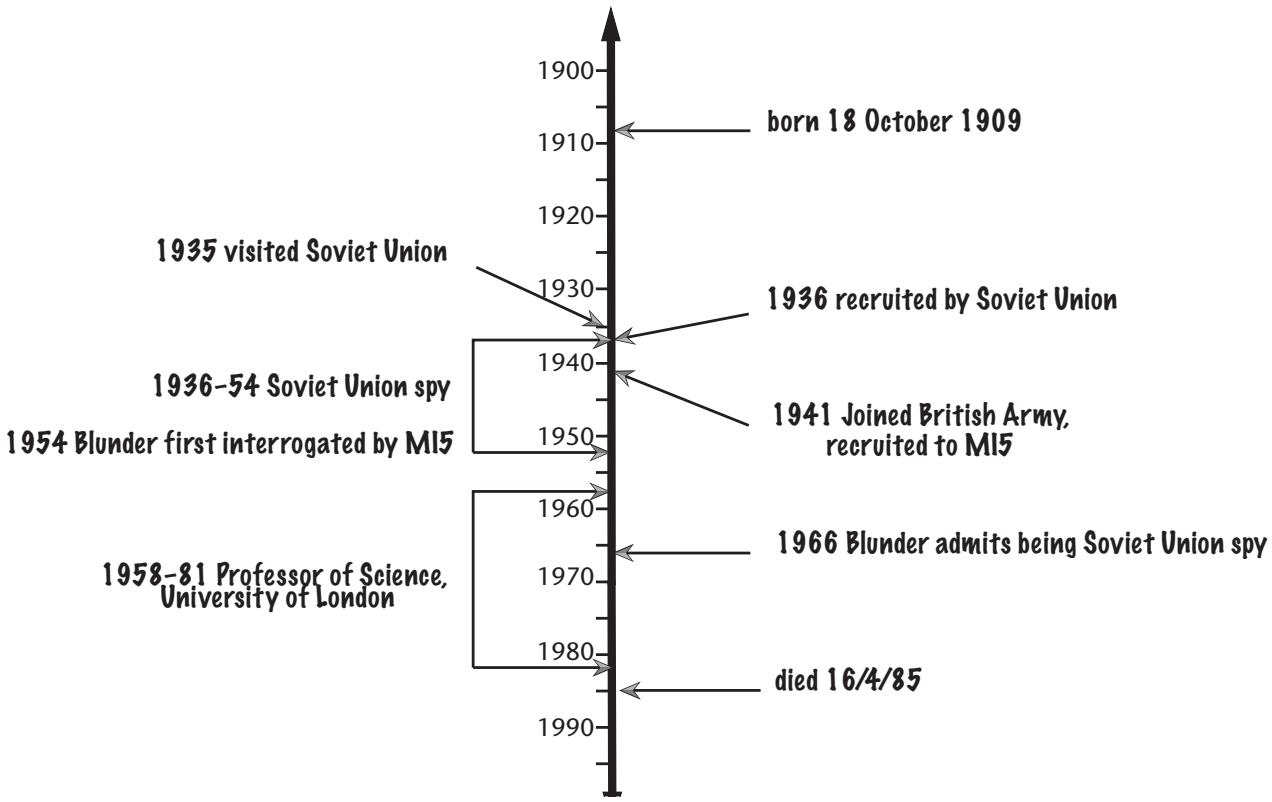
Espionage



A spy is a person employed to gather secret information or to spread false information. This work is known as espionage. Spies may be employed by governments to gain an advantage over their enemies, or by other organisations for commercial benefit. A spy may spend years building a 'normal' identity to hide his or her undercover activities and learn many techniques to secretly communicate information.

Watta B. Blunder was a British scientist who was exposed as a Soviet spy. A timeline of the main events in his life is shown below.



1 When was Blunder born?

2 When did he finish his work as Professor of Science at the University of London?

3 For how long was he a Professor of Science at the University of London?

4 For how long was he a Soviet Union spy?

5 When did Blunder visit the Soviet Union?

6 In what year was he recruited by the Soviets?

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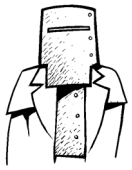

7 In what year was Blunder recruited to MI5, the British security service?

8 In what year did MI5 first interrogate Blunder?

9 In what year was Blunder finally exposed as a spy?

10 How old was Blunder when he died?

Have you ever sent a text message like 'wot r u doin 2nite?' If you have, then welcome to the world of secrecy, spies and espionage. Spies use complex codes and ciphers (a series of steps that decipher a certain code) to communicate secret information. Codes and ciphers have been around since the time of the Ancient Romans, and are one of the oldest forms of secret communication. Cryptography is the science of encoding and decoding secret messages.

11 Try to decipher the meaning of each of these sentences. Write the correct sentence and describe how you deciphered the code.

a tiphers hnd fodes nave keen qround gince rhe Yomans

b snamor eht ecnis sedoc desu evah seips

c th scnc f ncdng scrt mssgs s clld cryptgrphy.

The cipher used by Julius Caesar is rather simple. He changed the order of the letters in the alphabet by shifting them forward three places. His cipher looked like this.

Normal text	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Cipher text	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

12 Use the cipher to decode these messages.

a LI BRX ZLVK WR EO ORYHG, ORYH

b KH ZKR SURILWV EB FULPH, FRPPLWV LW.

The pigpen cipher is another form of substitution cipher. Symbols are substituted for the letters.

13 Use this cipher to decode the messages below.

a

b

14 Use wingdings font on the computer to create a cipher and write a secret message for your friend.

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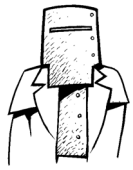
James Bond, 007



7

James Bond is a famous spy and is the main character in a large number of very popular action films. He is a fictional character created by Ian Fleming in 1953. Fleming wrote 12 novels and two short story collections featuring James Bond before he died in 1964. Other authors have since written James Bond stories.

EON Productions hold the rights to produce the Bond films and have made many successful movies starring this character. The film series has grossed over $4 billion. When this figure is adjusted for inflation it is almost $11 billion. This means the Bond movies are the highest grossing film series ever, outdoing even Harry Potter.

1 How many Bond movies have been produced so far?

2 Over how many years have the movies been produced?

3 What has been the longest break between making the movies?

4 Which movies were either side of the break?

5 Find the total of the box office for all the Bond movies to date, in billions of dollars.

6 Calculate the average amount earned by the films.

7 List the five top money earners in descending order.

8 List the five movies that earned the least amount in ascending order.

Title	Year	Box office (millions)
Dr No	1962	$425
From Russia with Love	1963	$556
Goldfinger	1964	$869
Thunderball	1965	$966
You Only Live Twice	1967	$720
On Her Majesty's Secret Service	1969	$513
Diamonds are Forever	1971	$618
Live and Let Die	1973	$786
The Man with the Golden Gun	1974	$427
The Spy who Loved Me	1977	$666
Moonraker	1979	$625
For Your Eyes Only	1981	$481
Octopussy	1983	$406
A View to Kill	1985	$316
The Living Daylights	1987	$363
Licence to Kill	1989	$272
Golden Eye	1995	$500
Tomorrow Never Dies	1997	$466
The World is Not Enough	1999	$505
Die Another Day	2002	$547
Casino Royale	2006	$641
Quantum of Solace	2008	$586

James Bond is renowned in the films for his gambling. Texas Holdem poker is the game played by Bond in the modern movies. A standard deck of 52 playing cards is used.

9 Calculate the probability of drawing from a standard deck:

- **a** the ace of hearts
- **b** a king
- **c** a heart
- **d** a heart or a diamond
- **e** an ace or a two
- **f** a jack or queen.

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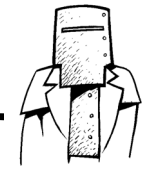
James Bond, 007 (cont'd)



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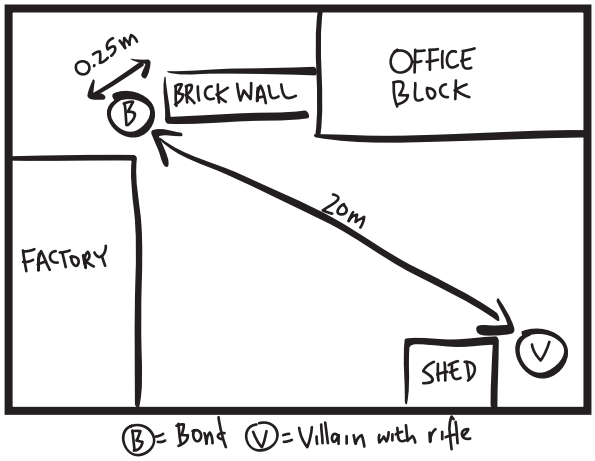


This is a storyboard sketch from one of the action scenes in a Bond movie. Does James Bond really have time to dodge a bullet from the villain's gun?

The villain has a rifle that has a muzzle velocity of 370 m/s.

The villain and James react at the same time.

Use the formula $t = \frac{d}{v}$, where t is the time in seconds, d is the distance in metres and v is the speed in m/s.

10 Calculate the time it takes for the bullet to reach its target.

11 Calculate the time it takes for James to duck for cover if he moves at a speed of 5.7 m/s.

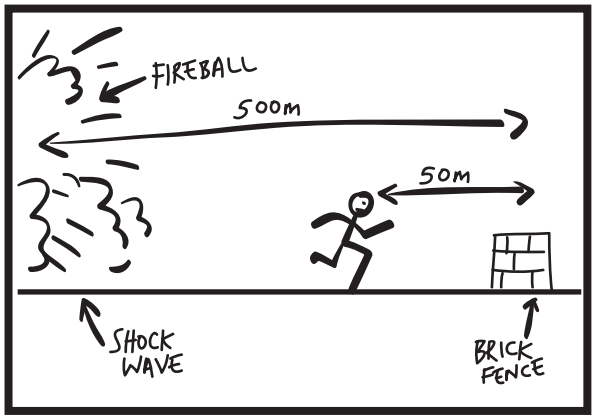
12 Did James have enough time to dodge the bullet?

13 Would James be able to dodge the bullet if he was 0.5 m from the brick wall?

14 What is the maximum distance James can move to be able to dodge the bullet?

15 Another villain in the same area has a 0.22 Rimfire rifle with a muzzle velocity of 1200 m/s. Is it possible for James to dodge a bullet from this rifle? Explain your answer.

Look at the storyboard sketch from another action scene. James is trying to outrun a fireball and shock wave. Can he reach safety behind the brick fence?



James can run at a speed of 8.2 m/s.

The speed of the fireball is about 400 m/s and shock waves can travel between 1200 m/s and 8200 m/s.

James begins to run as soon as the explosion occurs.

16 How long does it take James to reach the brick fence and safety?

17 Calculate the time it takes the firewall to reach the brick fence.

18 Calculate the time it takes the shock wave to reach the fence if it is travelling at 1200 m/s.

19 Was James able to reach the brick fence and safety in time?

20 Calculate the maximum distance James would need to be from the fence to be able to reach it safely.

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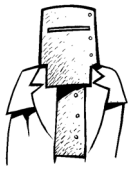
Lara Croft, Tomb Raider



8

Lara Croft is a fictional character and the star of the Tomb Raider video game series. As a thrill-seeking archaeologist her adventures take her into many situations involving criminal activities and espionage. Several movies have been produced that capture the exploits of Lara on the big screen.

When Lara Croft first appeared in the 1996 video game *Tomb Raider*, her three-dimensional character model was made of around 230 polygons. The model was later updated to add more realism by increasing the number of polygons to 4400. Further improvements saw this increased to 9800 and still further improvements took this to 32 800 polygons.

1 Calculate the percentage increase, to the nearest unit, in the number of polygons at each stage of improvement.

2 What was the largest percentage increase?

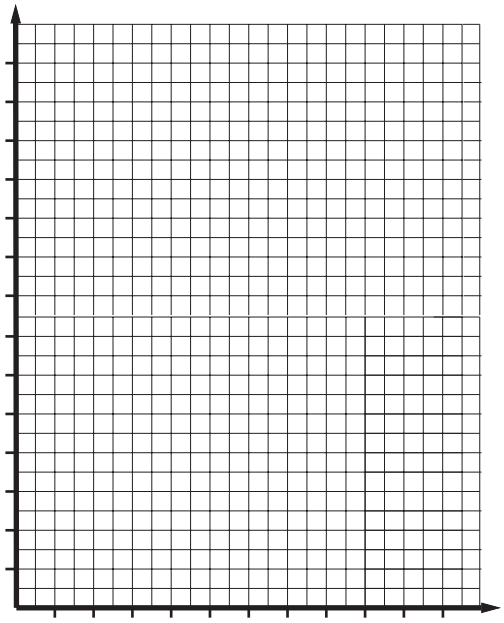
Fan web sites dedicated to Lara Croft appeared on the Internet after the release of *Tomb Raider*. More than 100 web sites were present by the end of 1998. By 2000 there were around 4700 web pages of Lara Croft stories on the Internet.

3 Calculate the percentage increase in the number of web pages devoted to Lara Croft.

One of the highlights of the movie *Lara Croft: Tomb Raider* was the 'bungee ballet'. During this sequence Lara's mansion is attacked by commandos seeking an artefact that she possesses. Lara needs to find out the correct length of cord she needs to make the drop in her bungee-harness device. She took these measurements during some trial jumps.

Length of cord (m)	5	10	15	20	25	30
Length of bungee drop (m)	12	28	29	54	61	65

4 On the axes provided draw a scatter plot of the data.



5 Draw a line of best fit through the points.

Use the graph to answer these questions.

6 Calculate the approximate bungee drop for cord lengths of:

a 7 m **b** 19 m **c** 4 m

d 32 m **e** 20 m.

7 What is the approximate length of bungee cord needed to complete a bungee jump of:

a 30 m? **b** 10 m? **c** 70 m?

d 65 m? **e** 50 m?

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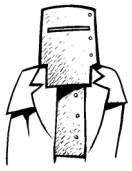
Lara Croft, Tomb Raider (cont'd)



8

In the movie *Tomb Raider: Cradle of Life*, Lara Croft is seen racing Terry along the Great Wall of China on a motorbike. The motorbike used is a TRX850 Yamaha, capable of speeds up to 215 km/h. Use the formula $v = \frac{d}{t}$ to assist with your calculations.

8 At a speed of 190 km/h, how long in hours would it take Lara to ride:

a 50 km? **b** 120 km? **c** 30 km? **d** 180 km?

9 If Lara is riding her bike at an average speed of 180 km/h, how far does she ride in:

a 3 h? **b** 4.2 h? **c** 1 h 30 min? **d** 45 min?

10 Complete the table showing the production costs and revenue of each movie.

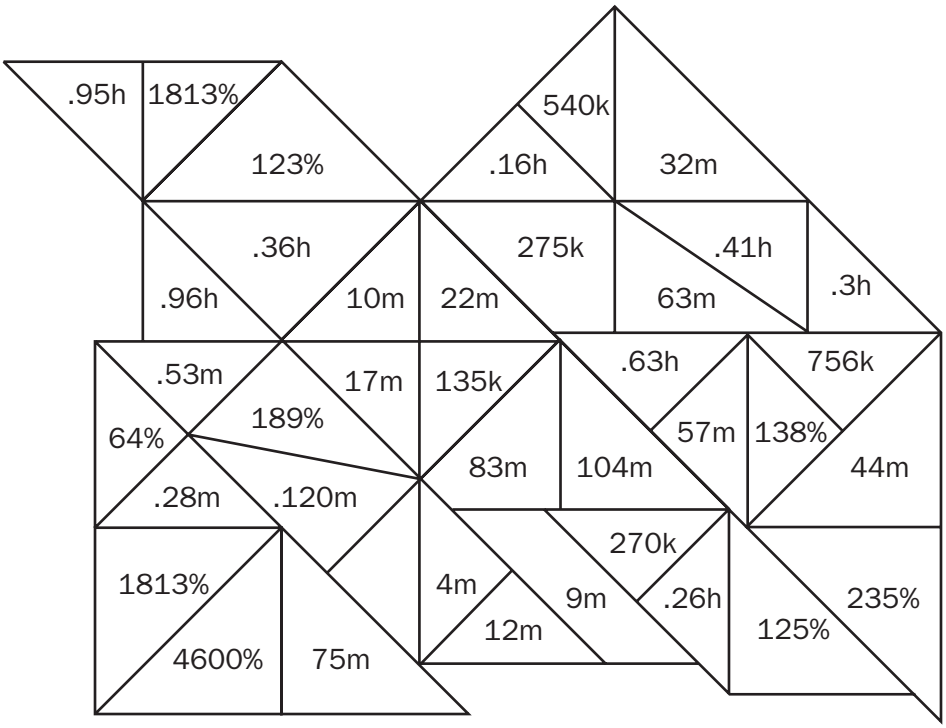
	Movie	Budget	Gross revenue	Gross profit	Gross % profit
a	*Lara Croft: Tomb Raider*	$115 million	$274 million		
b	*Tomb Raider: Cradle of Life*	$95 million	$156 million		

11 Which movie was more profitable? Why?

12 If the average cost of a movie ticket is $22, how many people saw:

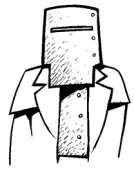
a *Lara Croft: Tomb Raider*? **b** *Tomb Raider: Cradle of Life*?

13 Find your answers to Q1–3 and 6–10 in the picture below. Colour the shape. Cut out the pieces and use them to form the picture of Lara on the run.



14 How many polygons did you use?

Time of death



9

Pathology is the study of disease and cause of death. A forensic pathologist specialises in examining dead bodies to determine how and when death occurred.

When a dead body is discovered, one of the first priorities is to determine the time of death. The pathologist starts by taking the temperature of the body. The normal temperature inside a human body is 37 °C. A pathologist will take the temperature inside the body, as well as the temperature of the location the body was discovered.

1 Write down some factors which may affect the rate at which a body cools down after death.

A formula used to predict the time since death is $t = -10\log_e \frac{T-21}{16}$, where t = time since death in hours and T is the current temperature of the body in °C. (The formula assumes that the air temperature is 21 °C and the temperature of the live body was 37 °C.)

2 Use the formula to calculate the time of death of these bodies. Complete the table.

	Time body found	Temperature of body	Time since death	Time of death
a	6.00 pm	36°C		
b	7.00 am	27°C		
c	6.15 am	30°C		
d	9.00 pm	22°C		
e	8.15 am	32°C		

3 Two suspects were observed near the scene of a crime. Willy was recorded as being there at 10 am and Marny at 11.30 am. A body was found at 12.15 pm and the pathologist measured its temperature to be 34 °C. Who is the more likely suspect? Why?

At 7.42 am on Sunday morning, Mr Costalotta was found dead by an isolated road in the country. The cause of death was determined to be a gunshot wound to the head. The day before the attack, three suspects were seen in the area: Jack Biggins at 10.00 pm, Tony Butcher at 5.30 pm and Bibby Brown at 11.45 pm. Your job is to determine the time of death using the sodium ion concentration and lysine levels in the body, and to identify the most likely suspect. At 9.00 am you measure the sodium ion concentration in blood serum to be 122.9 mM and the lysine levels to be 18 mM.

Table 1: Sodium ion concentrations following death.

Time (hours)	5	10	15	22	27	34	40	45	53
Sodium ion (mM)	127	126	124	121	119	116	114	112	109

Table 2: Lysine concentrations after death.

Time (Hours)	0	3	6	9	12	15	18	21	24	27	30
Lysine (mM/1000)	100	78	55	40	30	20	15	9	8	6	5

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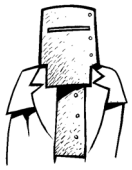
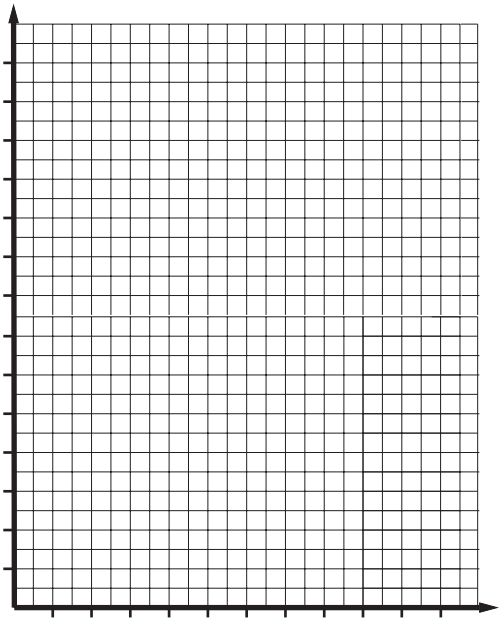
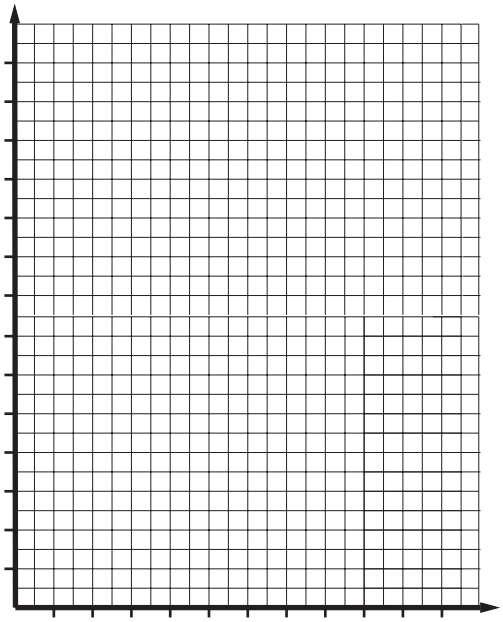
Time of death (cont'd)



4 Plot the graphs for both tables of data on the axes provided.





5 Use your graphs to estimate the time of death of the victim.

6 Who is the most likely suspect? Why?

Sodium		Average
Lysine		
Time of death		

The term 'stiff' is often used to refer to a dead body. It is a reference to the stiffness that occurs in body joints after death. This condition is known as rigor mortis and occurs because of the build up of wastes in the body. The table shows some of the changes in a dead body over time. Pathologists use this, and other information to estimate the time of death.

Time since death	Features of the body
1–2 hours	Early signs of lividity (patchy discolouration of the skin)
2–5 hours	Clear signs of lividity over the body
5–7 hours	Rigor mortis begins in the face
8–12 hours	Rigor mortis extends throughout the body
12 hours	Body has cooled to about 25°C internally
20–24 hours	Body has cooled to surrounding temperature
24 hours	Rigor mortis begins to disappear from the body in roughly the same order it appeared
36 hours	Rigor mortis has completely disappeared
48 hours	Body discolouration shows that putrification (rotting) has begun

7 These features were observed in a dead body. Estimate the time since death.

a A body has cooled to 25 °C.

b The face is stiff.

c There are no signs of rigor mortis.

8 How long does it take for rigor mortis to disappear from the body once it has set in?

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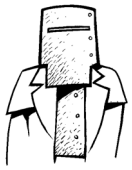
Who dunn it



10

Despite being a run-down filthy hovel, The Royal Oak is the only hotel in town. Tonight it has just five guests; each one is acting on behalf of the well renowned criminal Bugsy Moron. Their task is to rescue him from the local jail. From the clues, can you work out the real name and occupation of each guest, and the fake name, sometimes known as an assumed name, and occupation that each guest has used?

	Murderer	Robber	Cyber Thief	Marksman	Thief	Eric of Bondi	Bruno	George	Kelvin	Tyron the Terrible	Body guard	Bounty hunter	Priest	Road warden	Businessman
Damon															
Gareth															
Marcus															
Sheila															
Sandy Scarface															
Body guard															
Bounty hunter															
Businessman															
Priest															
Road warden															
Eric of Bondi															
Bruno															
George															
Kelvin															
Tyron the Terrible															

1 The robber who is posing as a wealthy businessman has not adopted the name of Kelvin or George.

2 The cyber thief has decided to call himself Bruno.

4 Sandy Scarface has disguised himself as a road warden.

5 The man posing as Tyron the Terrible, a body guard for hire, is not Sheila the marksman.

6 The murderer isn't pretending to be a bounty hunter.

Real name	Real occupation	Assumed name	Assumed role

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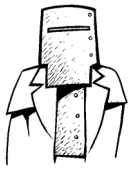
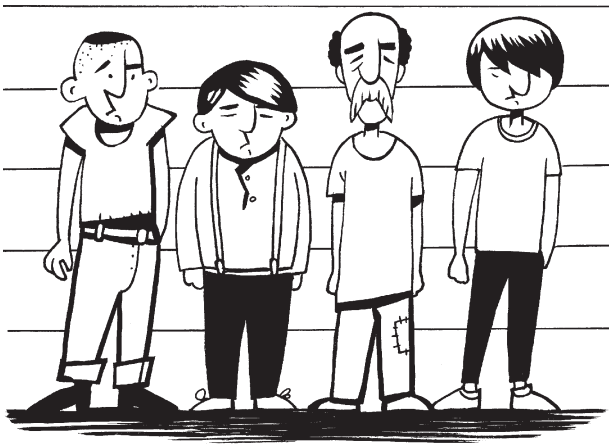
Who dunn it (cont'd)



10

The men in this puzzle were all sent to prison. When they were released from jail they started new careers. Using the clues, work out each man's real name, his offence and new career.

	Bartlett	Jefferson	Lemming	Brown	Chaming	Thief	Arsonist	Kidnapper	Fraudster	Vandal	Sheep farmer	Engineer	Baker	Wheat farmer	Shop owner
Alex															
Eril															
Charlie															
Peter															
Andrew															
Sheep farmer															
Engineer															
Baker															
Wheat farmer															
Shop owner															
Thief															
Arsonist															
Kidnapper															
Fraudster															
Vandal															



1. Alex went into baking after he got out of jail. Brown, who became a shop owner and managing director, was not Charlie.

2. Peter Lemming did not become a farmer; however, Eril, a kidnapper, did take up farming after leaving jail.

3. The arsonist became an engineer in a company that built ships and machinery.

4. Bartlett did not take up sheep farming.

5. The vandal, whose surname was not Chaming, became a farmer after leaving jail.

6. Jefferson went to jail after being convicted of theft.

First name	Surname	Offence	Current job

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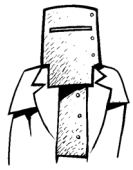
Motor vehicle crime



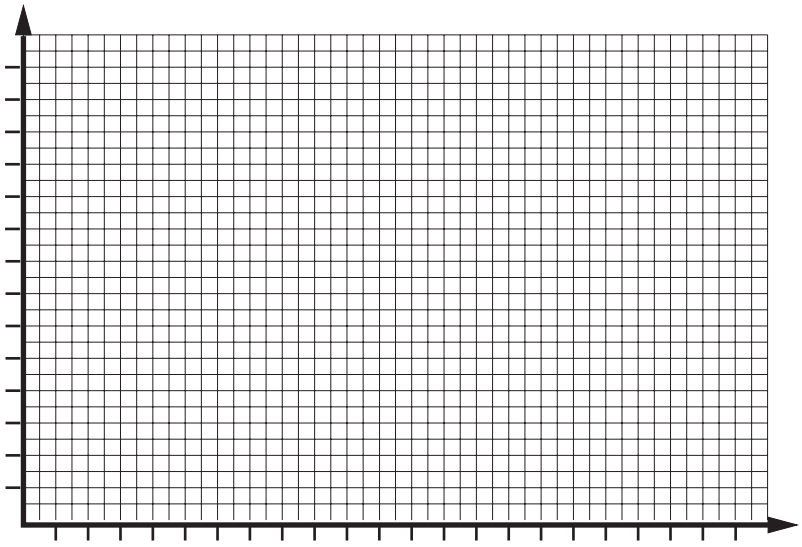
11

The theft of motor vehicles is a very common type of crime that affects everyone around the world. It is often linked to organised crime. The vehicles are not only stolen for their own sake; sometimes they are sold to finance or assist in other criminal activities.

The International Agency for Vehicle Crime has developed a database of records of reported stolen motor vehicles around the world. Over 100 countries share their national stolen vehicle records on the site and 150 countries use the database regularly.

1 On the axes provided, draw a column graph of the data in the table.

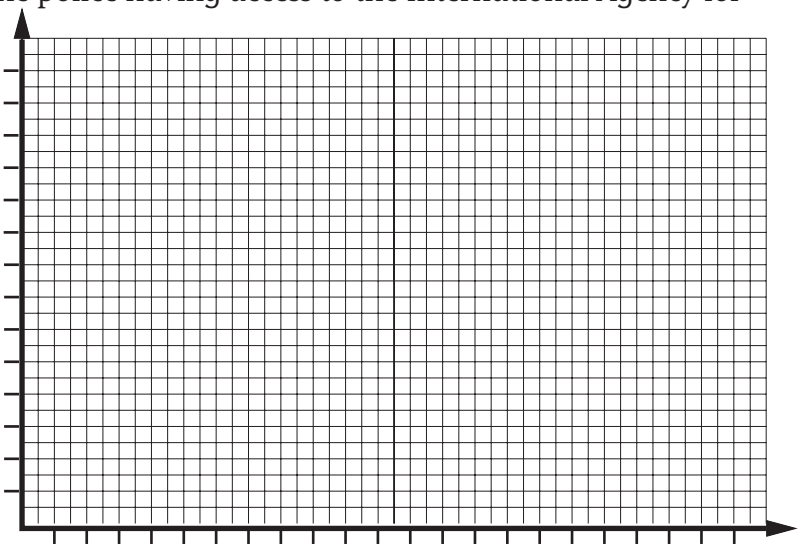
Year	Number of stolen vehicles (millions)
2000	2.3
2001	2.7
2002	2.6
2003	2.9
2004	3.1
2005	3.3
2006	3.7
2007	4.1
2008	4.5
2009	6.2
2010	7.0
2011	7.2



2 What are some reasons that might explain why there has been a large increase in the number of stolen cars registered on the database?

3 On the axes provided, draw a bar chart of the data in the table, which shows the number of stolen cars recovered due to the police having access to the International Agency for Vehicle Crime database.

Year	Number of stolen vehicles recovered
2000	12 000
2001	12 000
2002	15 000
2003	16 000
2004	12 000
2005	17 000
2006	34 000
2007	37 000
2008	31 500
2009	26 500
2010	33 000
2011	35 000



4 In which year were the most cars recovered?

5 In which year were the least cars recovered?

6 In which year were 12 800 cars recovered?

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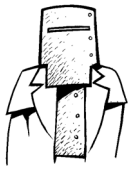
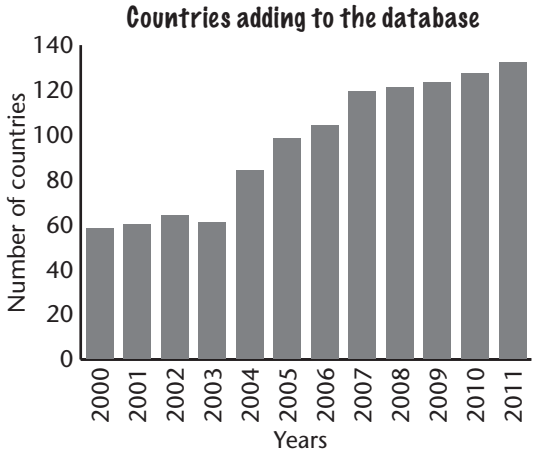
Motor vehicle crime (cont'd)



11

The graph shows the number of countries adding data to the International Agency for Vehicle Crime database of stolen cars.



7 Explain why there has been a gradual increase in the number of countries adding data.

8 Which two years have seen the largest increase?

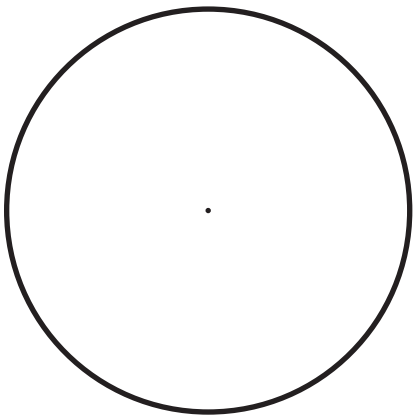
9 Which year saw a slight fall in the number of countries contributing?

10 In which year did 61 countries participate?

11 How many countries were participating in 2008?

	Year	Number of searches (×1000)	Sector angle
a	2000	777	
b	2001	861	
c	2002	1 359	
d	2003	1 709	
e	2004	1 776	
f	2005	2888	
g	2006	4 985	
h	2007	4 427	
i	2008	5 753	
j	2009	14 262	
k	2010	18 689	
l	2011	19 851	

12 Find the total number of searches on the database. Complete the table and draw a sector graph.



In one year there were approximately 11 million vehicles registered in Australia and approximately 110 000 motor vehicles reported to police as stolen.

13 How many cars per 100 000 registered vehicles were stolen?

14 Approximately 90 000 of the stolen vehicles were recovered. What is the percentage recovery rate?

15 This is a 20% decrease from the number of vehicles stolen in the previous year. How many vehicles were stolen in that year?

16 How many cars per 100 000 registered vehicles were stolen in the previous year?

Vehicle thefts occur from various locations. 40% of thefts occurred from footpaths and streets, 23% from residential locations, 12% from car parks and 10% from retail locations.

17 In that one year in Australia how many cars were stolen from:

a car parks? **b** footpaths and streets? **c** residential locations?

18 How many more thefts occurred from residential locations than retail locations?

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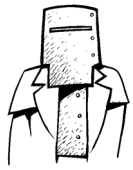
Hoon drivers



12

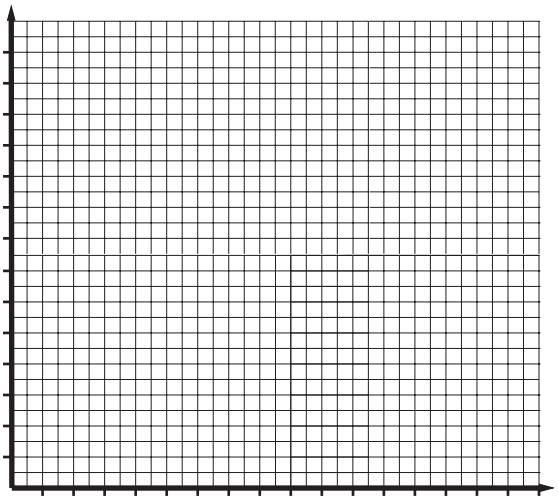
Many drivers cheat a little when driving. For example, they think that if the speed limit is 50 km/h, and they drive at 53 km/h they won't be pulled over by police. So they happily let the speedometer hover just above the speed limit, even though they are breaking the law and greatly increasing their chances of crashing.

1 The stopping distance for a car is based on the driver's reaction or thinking time to apply the brakes, and the actual braking distance for the car. Use the formulas below to calculate the total stopping distances, to one decimal place, for cars driving at various speeds.

Reaction distance = $0.208 \times v$ Braking distance = $0.025 \times v^2$

Speed v (km/h)	10	20	30	40	50	60	70	80	90	100	110
Reaction (m)	2.08										
Braking (m)	2.5										
Stopping (m)	4.58										

2 On the axes provided, plot speed against the three sets of distances on a single line graph. Use a different colour for each line on the graph.



3 At 90 km/h, how far does the car travel while the driver is thinking about braking?

4 Outside schools the speed limit is 40 km/h. What is the total stopping distance at this speed?

5 The speed limit in our streets was reduced from 60 km/h to 50 km/h. How many fewer metres does a car now need to stop?

6 How many metres more does a car need to stop when travelling at 100 km/h compared to 110 km/h?

7 Comment on the shape of each graph and what this means for drivers.

8 List three things that may increase a driver's reaction time.

9 List three things that may decrease the braking distance of a vehicle.

Hoon driving can include doing 'burn-outs', racing cars on public land, 'doughnuts' in car parks and repeated instances of excessive speeding or driving without a licence. The public are also encouraged to 'dob in a hoon'. Hoon drivers can have their vehicle impounded or, in repeat offences, lose their car for good.Within two years of anti-hoon laws being introduced in one state the police had impounded 4998 vehicles for hoon driving offences.

10 How many vehicles is this per day, on average?

11 Police claim they impounded 10 vehicles per day. What is the basis for this claim?

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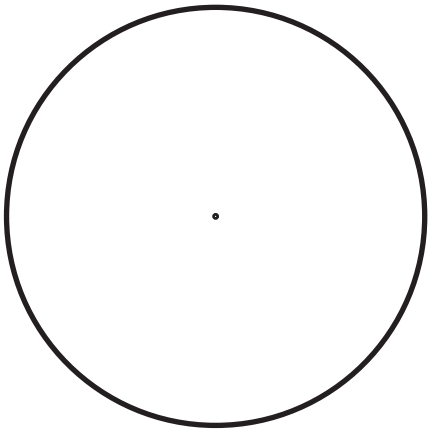
12 The table below shows a breakdown of 447 hoon offences. Complete the table, and then draw a sector graph.

	Offence	Number	Percent	Sector °
a	Burnouts	124		
b	Doughnuts	6		
c	Fishtails		13.4	
d	Monos on a motorbike	9		
e	Street racing	8		
f	Wheelies		8.6	
g	Speeding	195		
h	Driving dangerously		0.5	



13 437 of the 447 drivers were male. What percentage is this?

14 How many drivers were between 16 and 25 years of age, if this group committed 75% of the hoon offences?

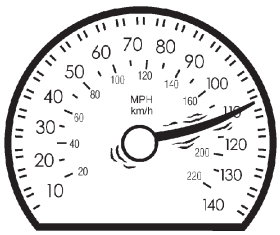
15 165 of the drivers were in the 36–40 years age group. What percentage is this?

16 Calculate the speed above the limit and the percentage above the limit for these offences.

a The highest speed detected was 220 km/h in a 60 km/h zone.

b The second highest speed detected was 200 km/h in a 60 km/h zone.

c The third highest speed was 187 km/h in a 110 km/h zone.



17 9% of the vehicles were motorcycles. How many was this?

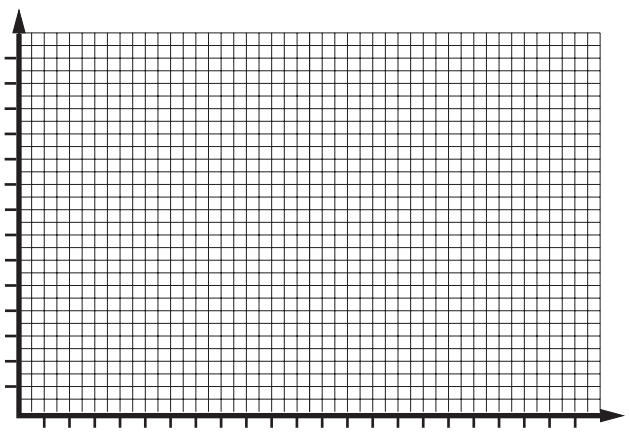
18 399 of the vehicles were cars. What percentage is this?

19 51% of the offences were committed in Commodores. How many cars is this?

20 1 offence was committed in a ute. What percentage is this?

21 The table below shows when the hoon offences were committed. Complete the table and plot a column graph of day and percentage.

	Day	Number	Percentage
a	Monday	41	
b	Tuesday		12
c	Wednesday		8
d	Thursday	54	
e	Friday	87	
f	Saturday		22
g	Sunday	76	



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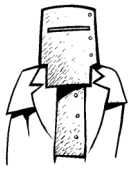
Crash scenes

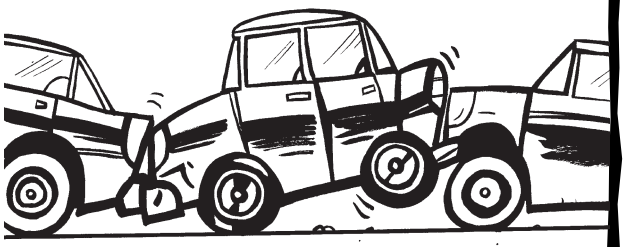


13

The scientific process of investigating, analysing and drawing conclusions about the causes and events during a vehicle collision is carried out by vehicular accident reconstruction. Re-constructionists try to identify the causes of the crash.

Forensic experts examine the skid marks at a crash scene as these can be used to determine an estimate for the speed that a vehicle was travelling. The formula used to calculate the speed of a car from skid to stop is given by:

$v = \sqrt{255\mu S}$, where v is the speed in km/h, S is the length of the skid mark in metres and μ is the coefficient of friction.



1 Calculate the speed that these cars were driving on a dry asphalt road, where $\mu = 0.7$ and the skid marks were measured to be:

a 25 m **b** 35 m **c** 45 m **d** 5002 cm **e** 20 m 35 cm.

2 Calculate the speed that these cars were driving at on a wet asphalt road, where $\mu = 0.4$ and the skid marks were measured to be:

a 20 m **b** 31 m **c** 34 m **d** 4 120 cm **e** 15 m 45 cm.

If you were an Indianapolis race driver, you would use 'slick' racing tyres with no tread. On dry surfaces you might get as high as 0.9 as a coefficient of friction, but driving them on wet roads would be dangerous since the wet road coefficient might be as low as 0.1.

3 Calculate the length of the skid mark an Indianapolis racing car would make on a dry asphalt track if it was travelling at:

a 160 km/h **b** 210 km/h **c** 260 km/h **d** 110 km/h **e** 190 km/h.

4 Calculate the length of the skid mark an Indianapolis racing car would make on a wet asphalt track if it was travelling at:

a 160 km/h **b** 210 km/h **c** 260 km/h **d** 110 km/h **e** 190 km/h.

5 Compare your answers for Q3 and Q4. What do you notice?

6 A driver claimed her car was travelling just on the speed limit in a school zone. She braked and stopped only centimetres from a student on a crossing. Investigators at the scene found that weather conditions were fine and sunny and the road surface was asphalt. The car's tyres were in excellent condition. They measured the skid marks of her car and found them to be 16 m long. Was she telling the truth? Explain your answer.

7 The driver of a Toyota Camry failed to give way to the right at a small intersection in a 50 km/h zone. A Commodore hit the side of the Camry with a considerable amount of force. He claimed, as part of his defence, that the other car was travelling significantly over the speed limit. Forensics on the scene found that the road was very wet, and the skid marks of the car that was allegedly speeding were 95 m long. The tyres of the Commodore were in poor condition and resembled the 'slicks' of racing cars. Does the driver of the Camry have any defence? Explain your answer.

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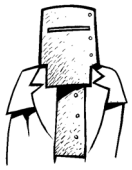
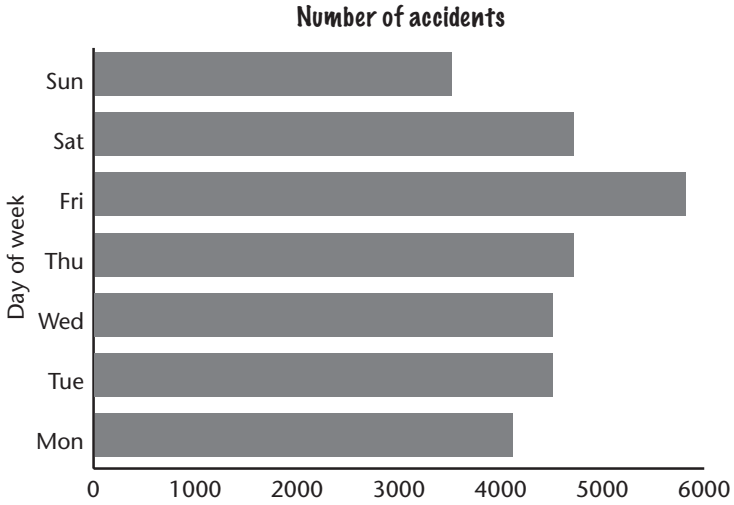


The following statistics are from one state's road safety report.

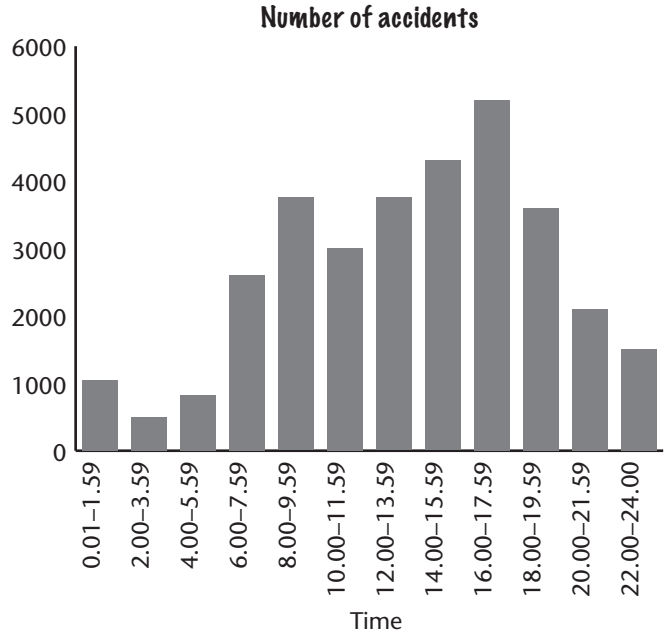
8 The graph opposite shows the number of car accidents occurring on different days of the week in the state.

- **a** Which day had the most accidents?
- **b** Which day had the fewest accidents?
- **c** How many accidents were there on Sunday?
- **d** Which day had approximately 4100 accidents?
- **e** Why do you think there are more accidents on Fridays?



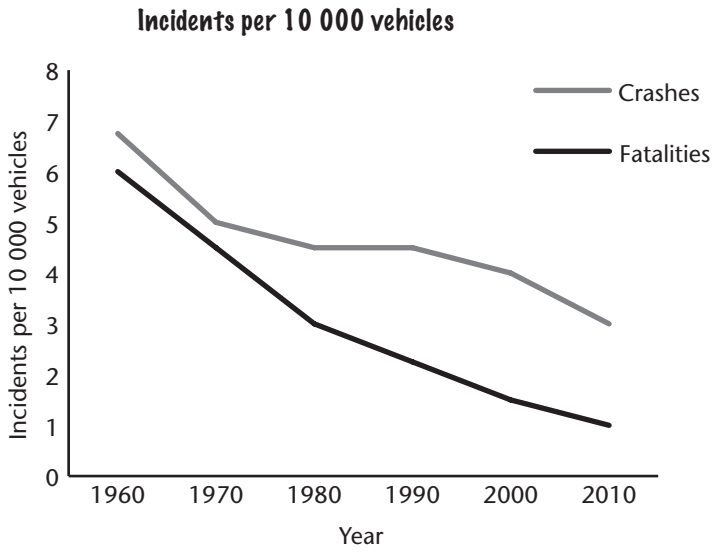
9 The graph opposite shows the number of accidents occurring at different times of the day.

- **a** Which time slot had most accidents?
- **b** Which time slot had least accidents?
- **c** Which time slot had 500 accidents?
- **d** How many accidents were in the 10.00–11.59 time slot?
- **e** Why do you think most accidents occur in the 4.00 pm to 5.59 pm time slot?



10 The graph opposite shows the changes in the rates of number of crashes and fatalities since 1960, per 10 000 vehicles.

- **a** Explain why there is a downward trend in both crashes and fatalities.
- **b** Why does the data use the rate per 10 000 vehicles, rather than the total number of incidents per year?



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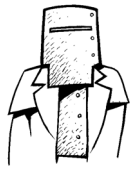
Blood stain analysis



14

Forensics uses many other disciplines of science to assist in solving criminal cases. Blood stain pattern analysis (BPA) is a specialty within the field of forensic science. Recent scientific and technological advances have improved the use and application of blood stains as evidence.

Blood is a tissue that is circulated within the body to assist other parts of the body. Blood has specialised cells that allow it to carry out its complex functions. In a healthy person, approximately 8% of their total weight is blood.

1 Calculate the volume of blood, in litres, in people that weigh: (Assume 1 g of blood is 1 mL)

a 70 kg **b** 45.3 kg **c** 63 kg 345 g **d** 56 kg 23 g

2 Calculate the weight, in kilograms, of a person with a volume of blood of:

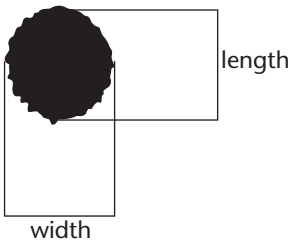
a 3.5 L **b** 3.56 L **c** 5617 mL **d** 8672 mL.

Blood contains three components suspended within plasma. The three components are erythrocytes, leukocytes and platelets. Plasma is a yellowish fluid composed of 92% water, 7% proteins and other materials such as salts and waste. Plasma makes up about 55% of blood. The remaining 45% is blood cells and platelets.

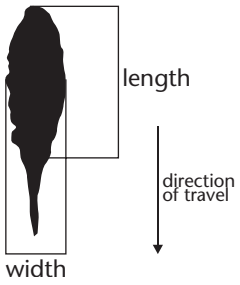
3 Complete the table.

	Volume of blood (L)	Volume of plasma to nearest mL	Volume of water to nearest mL	Volume of proteins to nearest mL
a	5.7			
b		2475		
c			1923	
d		2695		
e				293

A blood droplet in freefall has the shape of a sphere. If the droplet strikes a surface and produces a well-formed stain, it is possible to determine the angle at which this droplet struck the surface. The blood stain will be circular if it hits the surface at 90°. Blood that strikes the surface at an angle will have a teardrop shape.



90° impact



Angled impact

To calculate the angle of impact measure the width and length of the stain in millimetres, and use the formula $\text{angle} = \sin^{-1}\frac{\text{width}}{\text{length}}$. You will need a scientific calculator for this exercise.

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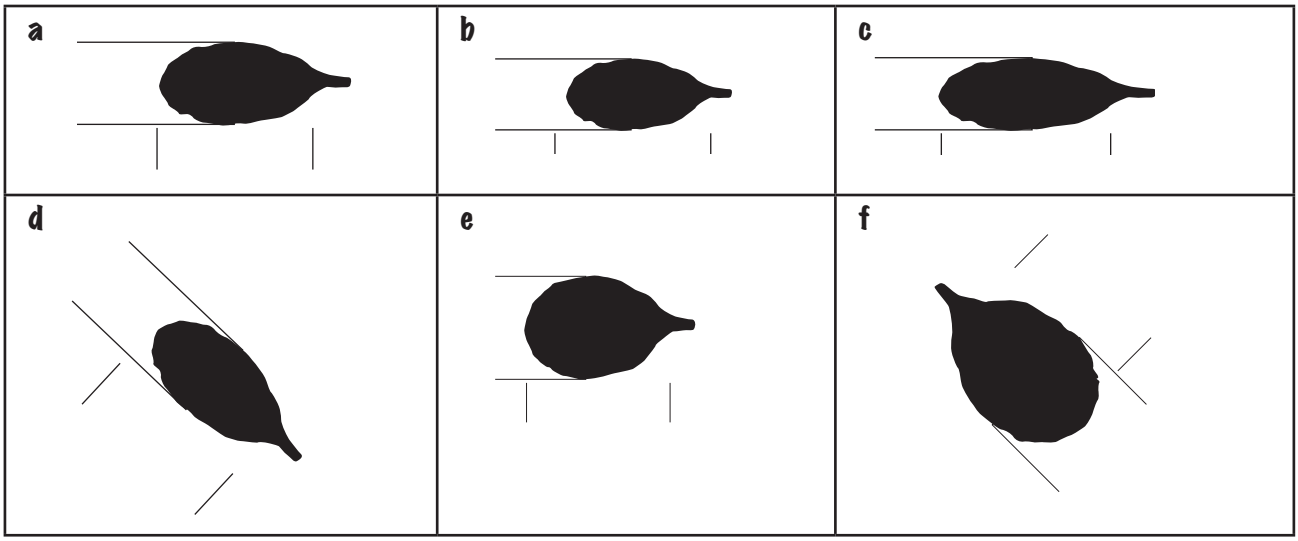
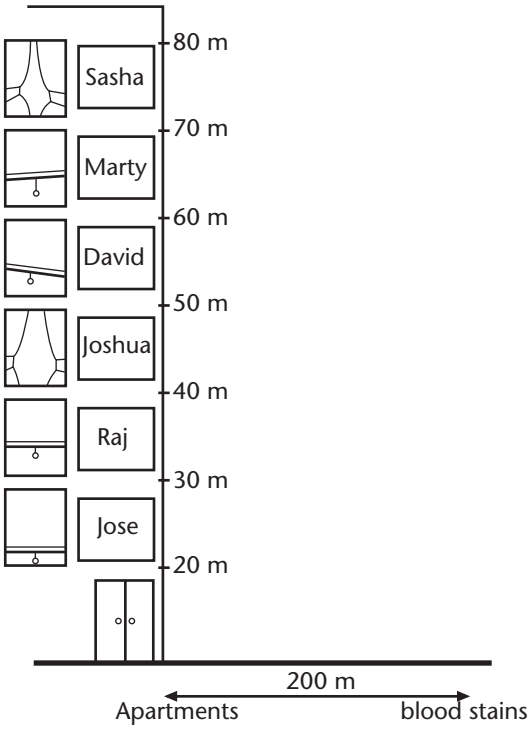
Blood stain analysis (cont'd)

4 Measure the length and width of each blood stain, then calculate the impact angle with the ground. State the direction of travel.



5 Forensic scientists discovered some blood stains at the scene of a crime and suspected shots had been fired from the building opposite. They took measurements of the blood stains and the building. Calculate the impact angle of each blood stain and the height the bullet was fired from to identify the suspect.



	Blood stain measurements		Impact angle	Building height	Possible suspect
	Width (mm)	Length (mm)			
a	3.6	10			
b	1	6.1			
c	3.8	12.3			
d	2.1	9.8			
e	3.25	11.8			

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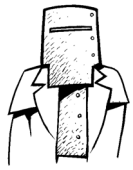
DNA



15

DNA or deoxyribonucleic acid is found in the centre of almost all living cells. DNA is an incredibly long, thin strand of material about two metres in length, coiled inside each cell. Forensic scientists use DNA, such as in a strand of hair found at a crime scene, to identify matching DNA of another individual, such as the suspect. This process is called genetic fingerprinting.

The following measurements were taken of the width of a DNA chain: 2.2, 2.4, 2.3, 2.2, 2.5 and 2.6 nanometres.

1 Calculate the average width of the DNA chain, in nanometres.

2 A nanometre is one-billionth of a metre, that is, 1×10^{-9} m. Express your answer to Q1 in metres:

a as a decimal

b in scientific notation.

A human DNA strand contains up to three billion base pairs, each pair being about 0.33 nanometres long.

3 Calculate the length of a human DNA strand. Write your answer in:

a nanometres

b metres.

4 The human body contains about 10^{14} cells. Each cell contains a DNA strand. Calculate the length of the entire DNA in the human body if it is lined up end to end. Write your answer in:

a nanometres **b** metres **c** kilometres.

5 How many times would this DNA reach the moon, if the average distance of the moon to Earth is 384 400 km?

6 How many times around the Earth would this DNA stretch if the diameter of the Earth is 12 576 km?

7 One of the great things about a DNA fingerprint is that it can be stored easily in computers. In the United States, the FBI has 5 million DNA profiles of convicts and over 200 000 profiles of DNA crime scenes. The population of the US is about 308 367 000. Calculate as a percentage of the total population the DNA profiles the FBI has:

a collected from convicts **b** of crime scenes.

8 In one country, 30 000 DNA samples are processed each year, each one taking just 5 days to analyse. If the population of this country is 34 036 000 people, calculate:

a the number of DNA samples processed per week

b the total amount of time, in years, it has taken to process these DNA samples.

9 In one European country the DNA database grows at about 400 000 per year. Everyone arrested even for a minor crime has their DNA profile analysed. The population of this country is around 62 784 500 people.

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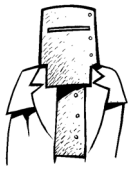

a Round off the population to the nearest million.

b Calculate the percentage growth per year of DNA profiles.

c At this rate, how long would it take to have a DNA profile for the entire population?

10 In another country, authorities claim that DNA profiles help identify suspicious individuals in 15 murder cases and 31 rapes every month.

a If there are 160 rapes per month, in what percentage of cases is DNA being used to identify the perpetrator?

b If there are 17 murders per month, in what percentage of cases is DNA being used to identify the murderer?

DNA is constructed using four different building bases: Adenine, Thymine, Cytosine and Guanine. DNA stores information about the individuals in a code created by the sequence of these bases. Much of the DNA appears to be junk. The length of these stretches of 'junk' DNA varies from person to person to form a genetic fingerprint of that person. Scientists use a process of electrophoresis to map the stretches of 'junk' DNA to produce a DNA profile, the result of which looks like a barcode found on supermarket goods.

11 Make a list of some common sources of DNA at a crime scene.

Matching DNA from a crime scene to DNA taken from a suspect is not an absolute guarantee of the suspect's guilt. Instead, forensic experts prefer to talk about probability of a DNA match. DNA is made up of four different bases: Adenine (A), Cytosine (C), Guanine (G) and Thymine (T).

12 Suppose that the bases are randomly distributed along a single strand of the DNA. For example, if a DNA single strand is 3 bases in length, what is the probability that it contains only one A base?

A	N	N
$\frac{1}{4}$	$\frac{3}{4}$	$\frac{3}{4}$

N	A	N
$\frac{3}{4}$	$\frac{1}{4}$	$\frac{3}{4}$

N	N	A
$\frac{3}{4}$	$\frac{3}{4}$	$\frac{1}{4}$

There are three different arrangements (the N means 'not Adenine'), so

$$\Pr(\text{A}) = 3\left(\tfrac{1}{4}=\tfrac{3}{4}=\tfrac{3}{4}\right)= \tfrac{27}{64}= 0.42$$

a If a DNA strand is 4 bases in length, what is the probability that it contains only one T base?

b If a DNA strand is 5 bases in length, what is the probability that it contains only one G base?

c If a DNA strand is 4 bases in length, what is the probability that it contains two C bases?

d If a DNA strand is 5 bases in length, what is the probability that it contains only two A bases?

e The human genome is approximated 6 billion bases in length. What is the probability that another individual has the same genetic composition as you?

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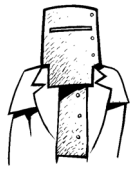
Fingerprints



16

Experts have come to believe that each person's fingerprints are unique, even those of identical twins. The probability of two people having the same fingerprint is more than one in sixty billion.

Experts start comparing fingerprints by examining the core of the prints. The core can be one of three patterns: arch, loop or whorl. Once they have identified the basic core type of fingerprint, the characteristics (minutiae) of the ridges are compared. Some of these minutiae are shown in the table below.

Ridge ending	Dot	Lake	Bifurcation	Spur	Cross over	Trifurcation

1 Identify some of these ridge characteristics in the fingerprints below. Circle and label each feature you find.

Experts have calculated that the probability of any two people having the same characteristic in the same spot is approximately $\frac{1}{10}$ for a ridge ending or bifurcation and between $\frac{1}{20}$ and $\frac{1}{100}$ for the other features.

2 Assume that the probability of two people have the same characteristics is 1/10 for each feature. Using the multiplication law of probability calculate the probability of two people having:

- **a** two characteristics in corresponding locations
- **b** three characteristics in corresponding locations
- **c** four characteristics in corresponding locations
- **d** five characteristics in common.

3 The population of Victoria is almost six million people. How many characteristics would have to be considered so that the probability of two people having this number of characteristics is less than $\frac{1}{16\,000\,000}$?

4 If one characteristic of a suspect's fingerprint matches a fingerprint found at the scene of the crime, what can the experts conclude?

5 If there are eight characteristics of a suspect's fingerprint that match a fingerprint found at the scene of a crime, fingerprint experts will be confident the suspect was at the scene. Why is this?

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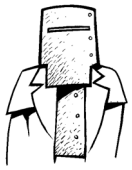

6 Below are images of two fingerprints. Identify and match corresponding minutiae and calculate the probability of the fingerprints being the same.

7 This fingerprint is a copy of a fingerprint found at the scene of a crime.

The fingerprints below are from several suspects. Your task is to examine the fingerprints to find out if one of the suspects was at the scene of the crime, and to calculate the probability of the suspect being at the scene.

Charger Collins Bulky Billson Dagger Duggan

The suspect is most likely to be _____.

8 The FBI need 15 matching minutiae to conclude that two fingerprints are identical. What is this probability?

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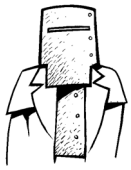
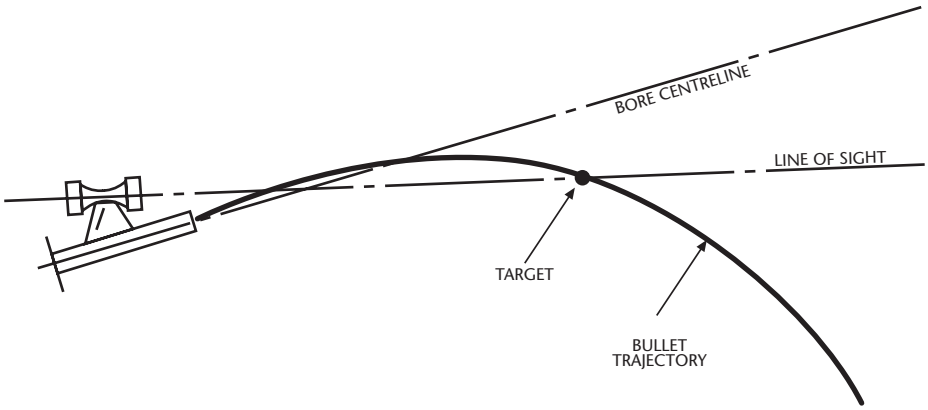
Ballistics



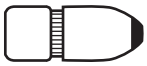
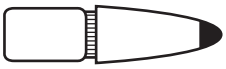
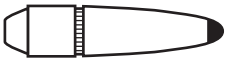
17

A bullet is a projectile propelled by a rifle or handgun. Many people believe that bullets fly in a straight line. This is untrue, they travel in a parabolic trajectory – one that becomes more and more curved as the distance the bullet travels increases. The centre line of the bore (the inner diameter of the gun barrel) is angled slightly upward in relation to the line of the sights (which are above the bore) so that the projectile crosses the line of sight on its way up and again on its way down.

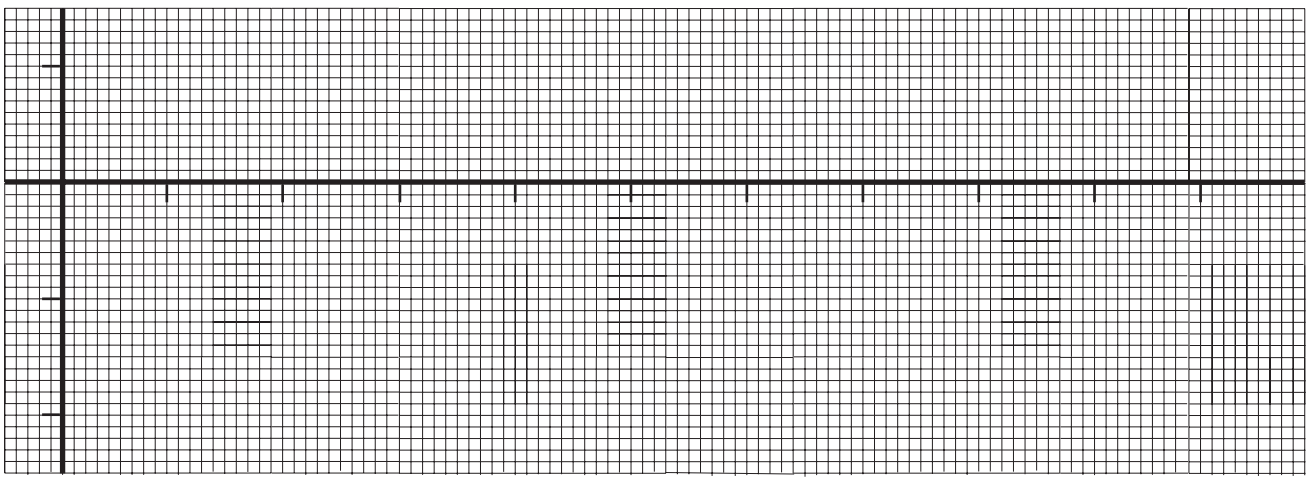
The Ballistic Coefficient (BC) is basically a measure of how streamlined a bullet is; that is, how well it cuts through the air. The higher the BC number the less drag and the better it cuts through the air.



The table below contains measurements of the trajectory from the line of sight (in mm) of three 6 mm bullets of the same weight but different shapes.

Bullet type		Ballistic coefficient	Distance from muzzle (m)					
			0	90	180	270	360	450
Round nose bullet		0.216	-38	+122	+121	0	-404	-1168
Spire point		0.357	-38	+97	+119	0	-283	-775
Spire point boat tail		0.400	-38	+92	+113	0	-264	-726

1 Plot the graphs of the trajectory for each bullet.



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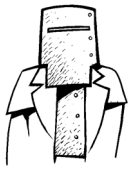


2 What is the distance at which each bullet passes through the line of sight?

3 Which bullet type would you use to shoot at long range targets? Why?

A Ballistic coefficient has units of lb/in² or kg/m². Normally BCs are stated in lb/in² by gun projectiles producers without referring to this unit. The formula for calculating the ballistic coefficient for bullets only is as follows: $BC_{bullet} = \frac{M}{i=d^2}$

Where:

M is the bullet mass

i = the form factor and

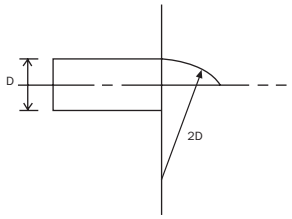
d = the diameter of the bullet.

	BC (lb/in²)	*M* (lb)	*i*	*d* (in)
a		0.0114	0.554	0.22
b		0.0164	0.501	0.24
c	0.540		0.511	0.24
d	0.595		0.482	0.26
e	0.588	0.0203		0.26
f	0.599	0.0231		0.28
g	0.527	0.0271		0.3
h	0.570	0.0271	0.502	
i	0.760	0.0429	0.494	

4 Use the formula to complete the table.

Spitzers or tangential ogive bullets are constructed on paper by drawing two arcs with a compass, the radius of each being identical, and being specified in calibres. The two arcs are drawn so that they start at the shank and curve toward the centre line of the bullet, crossing or at least touching the centre line. The origin of the radius is located on a line perpendicular to the centreline and even with the end of the shank. A spitzer is designated in calibres of radius, such as 2-S, 6-S or 10-S. This means that the length of the radius of the ogive arc is 2, 6 or 10 calibres.

5 Use a compass to construct Spitzer bullets of 2-S, 6-S and 10-S designations for each of these bullets.

2-S	6-S	10-S
		

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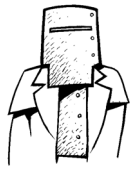
Graffiti



18

Graffiti is the act of marking a property or wall, without the consent of the owner, with writing, symbols or drawings. Graffiti is illegal and is a persistent problem in almost all towns and cities. The main motivators for graffiti artists are fame, adrenaline rush, artistic expression and malicious damage.

In one year approximately 14 000 incidents of graffiti were reported to police.

1 Calculate the number of incidents at each location if:

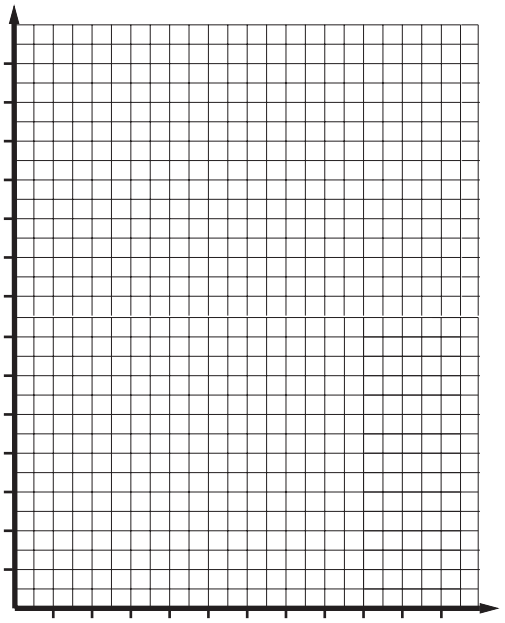
a twenty-six percent took place at residential properties

b twenty-five percent took place at educational premises

A breakdown of the locations of 14 000 acts of graffiti vandalism is shown in the table.

2 Draw a column graph to display the data in the table.

Location	Number of crimes
Business	2560
Carpark	480
Education	3500
Health	98
Industrial	99
Outdoor/public	1900
Public transport	758
Recreation	420
Residential	3640
Utilities	180
Religious	250
Hotels	140
Police	75



3 What percentage of the graffiti was committed in:

a car parks?

b religious buildings?

c public transport?

d business buildings?

In some places the act of vandalising by graffiti has some large penalties. The penalty could be $2200, 6 months in prison or a period of community work. Possession of spray paint can also lead to a fine of $1100 or 3 months in prison or a period of community work.

4 Write the 2 fines in thousands of dollars.

5 Write the terms for imprisonment in:

a weeks

b days (assume there are 30 days in a month).

The estimated cost of cleaning graffiti in Australia is about $400 million per year.

6 If the population of Australia is about 21 875 000 people, how much money is spent on graffiti clean-up per person?

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7 The cost of maintaining a hospital bed is approximately $1100 per day. How many hospital beds could be maintained in Australia per year with the graffiti clean-up money?

8 The cost of maintaining an aged care bed is approximately $100 per day. How many aged care beds could be maintained in Australia per year with the graffiti clean-up money?

9 What is your reaction to the answers above?

In one neighbourhood survey twenty-six percent of those questioned identified graffiti and vandalism as the third most important problem in their neighbourhood, after dangerous driving and theft from homes.

10 How many people in the state would this be if the state had the following populations?

a 5 443 000

b 7 134 000

c 2 176 900

Graffiti in a neighbourhood may be a sign that the police, local councils and residents are not in control or not concerned about the area. This may encourage further criminal activity which leads to economic decline and social instability. Much research has been conducted into this issue.

11 Researchers conducted an experiment where they observed 56 people passing by a letterbox. In a no-litter and no-graffiti situation, 8 people took the $20 note sticking out from the letterbox. What percentage of passers-by is this?

12 When the same letterbox was graffitied and litter was lying around, 13 out of 48 passers-by took the $20 note. What percentage of passers-by is this?

13 Researchers also found that of 80 people using an alley, 55 people littered in the graffitied area and only 26 people littered in the non-graffitied alley. Calculate the percentage of people littering in the:

a graffitied alley **b** non-graffitied alley.

A council removed approximately 411 400 incidents of graffiti from local government areas in one year. The graffiti covered almost 170 000 m^2.

14 On average, what is the surface area of a graffiti incident?

15 A can of spray paint covers approximately 0.9 m^2. How many cans of spray paint have the graffiti artists used?

16 If each can of spray paint costs $2.50, how much has been spent on applying the graffiti?

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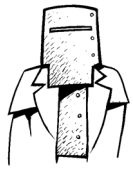
Cyberbullying



19

Cyberbullying is when a person is harassed or threatened by another person or group of people using technology like the Internet and mobile phones. Young people are particularly vulnerable to this type of hostile behaviour. They may receive text or email messages that contain hurtful and embarrassing comments or threats. Cyberbullies may also use social networking sites to start gossip or disclose private information about another person.

A research study discovered that over 60% of students were victims of cyberbullying.

1 How many students have probably been bullied in schools with these numbers of students?

a 350 **b** 249 **c** 562 **d** 491 **e** 1 265

Q2–6 are based on another survey of 1500 students.

2 Forty-two percent of the students had been bullied while online, and of the students who had been bullied one in four has been bullied more than once. How many students have been:

a bullied?

b bullied more than once?

3 Thirty-five percent have been threatened online, and of those threatened one in five have been threatened more than once. How many students have been:

a threatened?

b threatened, more than once?

4 Twenty-one percent of students have received mean or threatening emails or other messages. How many students is this?

5 Fifty-eight percent of students admit someone has said hurtful things to them online. Of those 58% students, more than four out of ten said it happened more than once. How many students have received:

a hurtful messages online? **b** hurtful messages more than once?

6 Fifty-six percent of students have not told their parents they have received hurtful messages online. How many students is this?

7 Repeat the calculations for Q2–6 based on the number of students in your classroom. Therefore, determine the statistical extent of cyberbullying in your classroom.

8 Discuss how big an issue cyberbullying is for students based on your calculations for Q7.

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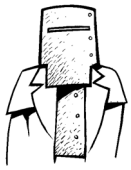
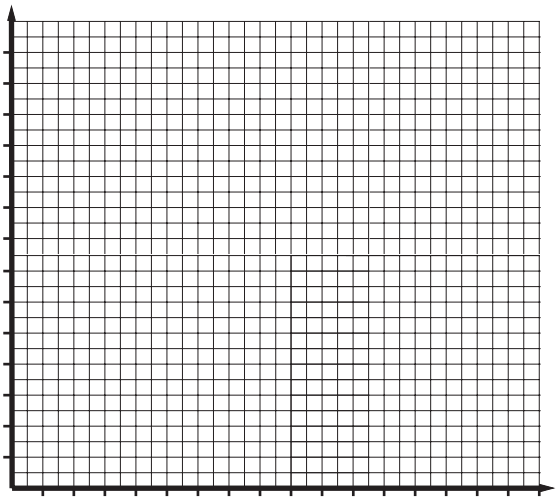


9 The table below contains data related to cyberbullying. Draw a column graph to display the data.

Technology used to bully	Percent
Email	23
Chat rooms	35
Text messages on mobiles	41

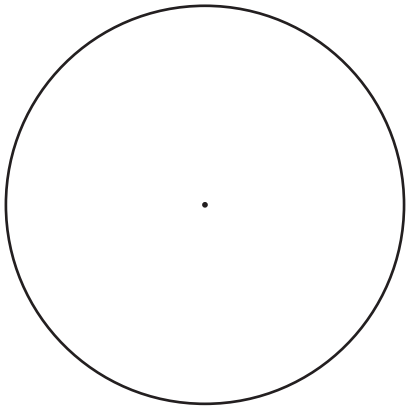


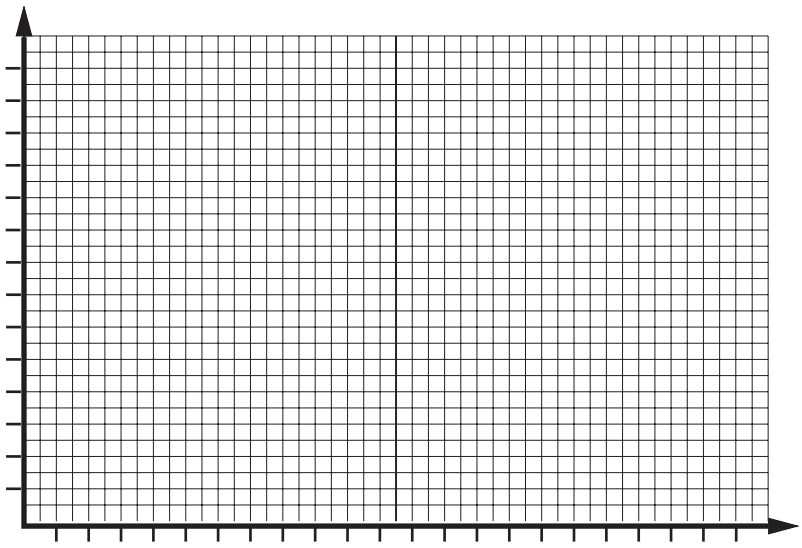
10 Another study of approximately 1500 Internet-using adolescents found that over $\frac{1}{3}$ reported being victimised online, and over 16% of respondents admitted to cyberbullying others.

a How many reported being victimised?
b How many adolescents bullied others?

11 Represent the following data using a sector graph

	Nature of internet bullying	Percentage	Sector angle
a	Disrespected	41	
b	Called names	19	
c	Physically threatened	12	
d	Scared for their safety	5	
e	Other minor incidents	23	





Another survey of 770 youths between the ages of 11 and 19 years found that many youths are not comfortable telling an authority figure about being a victim of cyberbullying for fear their access to technology will be taken from them. The study reported that while 24% told a parent, 14% told a teacher, 28% did not tell anyone and 41% told a friend.

12 How many students told a:

a teacher? **b** friend? **c** no one?

13 Represent the number of youths, using a horizontal bar graph on the grid above.

14 Discuss whether you think it is important to tell someone if you are being cyberbullied.

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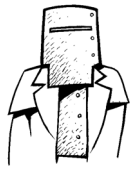
Environmental crime



20

The world is now facing numerous environmental challenges and threats. Environmental crime, including illegal trading of natural resources and dumping of hazardous waste, is a serious and growing international problem. As the world's natural resources rapidly diminish, environmental crimes such as illegal logging and fishing are increasing and are billion dollar activities.

In 1989 the oil tanker *Exxon Valdez* ran aground on Bligh Reef, spewing vast amounts of oil into the ocean.

1 The tanker carried almost 240 million litres of oil. Fifty million litres of this oil spilled into the ocean.

- **a** What was the fraction of oil that spilled into the ocean?
- **b** What is this fraction as a percentage?
- **c** What percentage of oil did not spill into the ocean?

2 Alaska has a coastline of approximately 6000 miles.

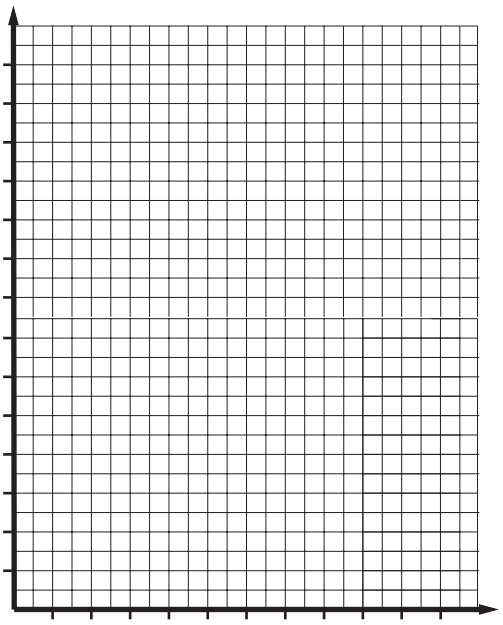
- **a** How many kilometres is this if 1 mile is the same as 1.61 kilometres?
- **b** 1300 miles of the shoreline was polluted by oil. How many kilometres is this?
- **c** What fraction of the coastline was polluted?
- **d** What percentage of the coastline was not polluted?

3 Approximately 35 000 birds and 1000 sea otters were confirmed to be dead after the spill.

- **a** What is the total number of dead animals found?
- **b** What fraction were sea otters?
- **c** What percentage were birds?

4 Since most of the dead sank into the water, it was estimated that the total number killed was 250 000 seabirds, 2800 sea otters, 300 harbor seals, 250 bald eagles, 22 killer whales, and billions of salmon and herring eggs. Disregarding the fish, make estimates based on the figures above of the:

- **a** total number of birds and animals killed
- **b** fraction (decimal) of sea birds killed
- **c** fraction of sea otters killed
- **d** percentage of harbour seals killed
- **e** percentage of birds killed.
- **f** Draw a column graph on the axes provided to display the loss of animals (do not include sea birds).



5 Exxon was fined $150 million. Since Exxon cooperated in the massive clean-up and also paid private claims, the court forgave $125 million. The remaining $25 million was split between the North American Wetlands Conservation Fund and the national Victims of Crime Fund.

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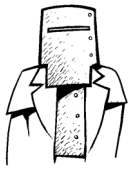

a What is the decimal fraction for the amount the court forgave Exxon?

b What is this amount as a percentage?

6 Exxon also paid $100 million to federal and state governments for environmental projects. Exxon also agreed to a $900 million settlement in a civil agreement. The $900 million went to pay for further environmental projects.

a Find the total amount, in billions of dollars, that Exxon had to pay for the oil spill. Don't forget the fines.

b What fraction was given to federal and state governments for environmental projects?

Serious organised crime in the forestry and timber industries is one of the most pressing environmental issues facing the global community. Significant volumes of natural resources are harvested, sold, purchased, transported or exported using illegal methods. The proceeds of illegal logging are laundered in the Asia-Pacific region, fostering corruption and undermining environmental governance.

7 Illegal forestry is directly linked to climate change as around one-fifth of global greenhouse gas emissions are linked to forest loss.

a Express the amount of illegal forestry linked to climate change as a decimal fraction.

b What is this fraction as a percentage?

8 Indonesia has the highest rate of deforestation in the world, at almost two million hectares each year. How many hectares is this per:

a month? **b** week? **c** day? **d** hour?

9 A football field is approximately 2 hectares in size. How many football fields of forest are being cleared per hour in Indonesia?

10 Satellite images show that 60 million hectares of Indonesia's forests are in a severely damaged state. How many equivalent football fields of forest is this?

11 A recent report has revealed that illegal logging is taking place in 37 out of Indonesia's 41 national parks.

a Express the number of national parks not being logged as a decimal fraction.

b What is the percentage of parks being logged?

12 Reports state that predatory loggers in Papua are illegally smuggling up to 300 000 cubic metres of merbau logs a month to China's booming wooden flooring industry. If a merbau tree produces 5 cubic metres of timber, how many merbau trees are being logged in a:

a month? **b** day (assume 30 days per month)?

c year? **d** week?

13 It is estimated that the local community in Papua receives $11 per cubic metre for the timber, the Papuan exporter receives $120 and the Chinese importer receives $240. The manufactured price in China is $468 and the retail price in imported countries such as the UK, US and Australia is $2300. Calculate the total cost paid per cubic metre, then the percentage amount at each stage.

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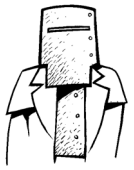
The cost of crime



Crime is costly to individuals and society as a whole. Costs include medical treatment and lost productivity from victims of assaults and homicides, property lost in burglaries and car thefts, credit card and employee fraud, arson and vandalism.

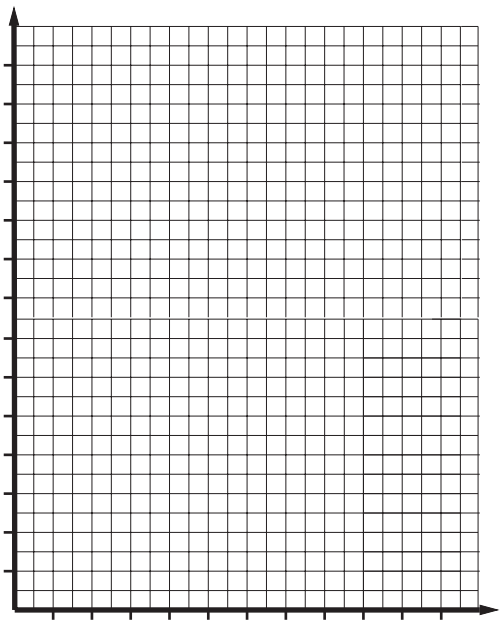
A study was undertaken by a regional government into the cost of crime in their state.

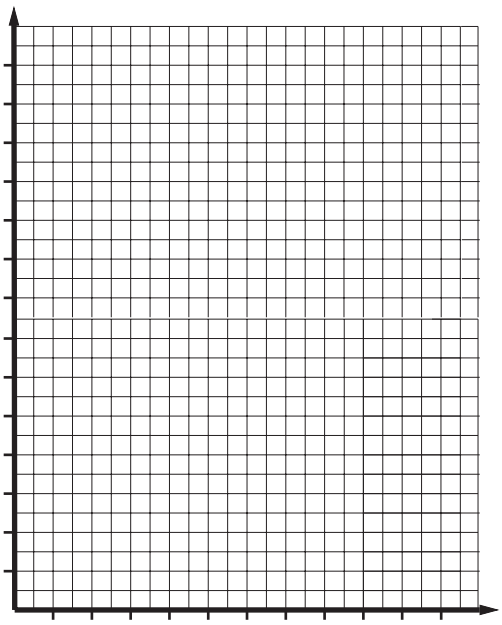
1 Round off the cost to two significant figures, complete the table and draw a column graph of the data.

	Type of crime	Cost ($millions)	Rounded cost
a	Homicide	240	
b	Assault	353	
c	Sexual assault	180	
d	Burglary	559	
e	Robbery	57	
f	Vehicle theft	150	
g	Fraud	2129	
h	Theft from vehicles	130	
i	Shop theft	213	
j	Other theft	74	
k	Arson	416	
l	Criminal damage	393	
m	Illicit drug harm	459	

2 Calculate the total cost of crime. Complete the table and draw a horizontal bar graph of the data.

	Type of crime	Cost ($millions)	Percent cost
a	Homicide	240	
b	Assault	353	
c	Sexual assault	180	
d	Burglary	559	
e	Robbery	57	
f	Vehicle theft	150	
g	Fraud	2129	
h	Theft from vehicles	130	
i	Shop theft	213	
j	Other theft	74	
k	Arson	416	
l	Criminal damage	393	
m	Illicit drug harm	459	
	Total		





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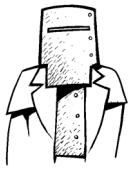
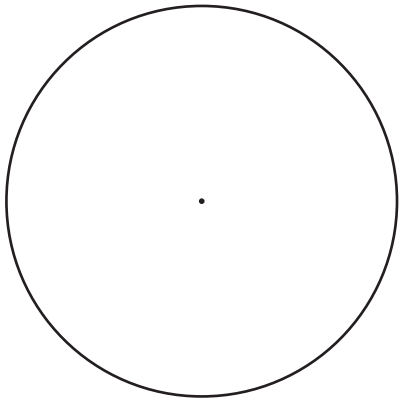
The cost of crime (cont'd)



3 The table below shows some other costs associated with crime in the same state. Find the percentage amounts and the sector angle to draw a sector graph of the data.

	Other costs	Cost ($millions)	Sector angle
a	Prosecutions	2450	
b	Victim support	268	
c	Law enforcement	748	
d	Insurance	145	
e	Total		



4 The horizontal bar graph shows the percentage breakdown of the numbers of crimes committed and the cost of each crime.

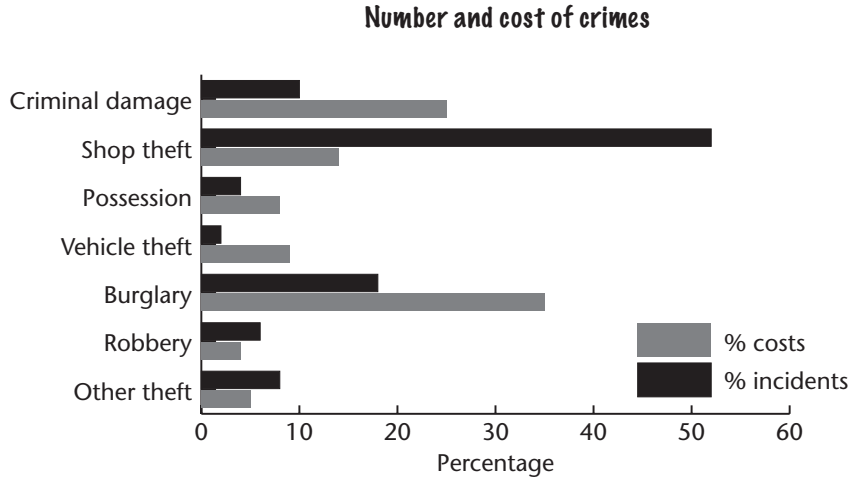
a Which type of crime is most frequently committed?

b Which type of crime is least likely to be committed?

c Which crime costs the community the most?

d Which crime costs the community the least?

e Which crime contributes approximately 25% of the costs?



f Which crimes represent approximately 6% of the incidents?

5 It is estimated that the cost of crime in Australia was $36 billion in one year alone. The amount spent on education in that year was $47 billion and the amount spent on health was $84 billion.

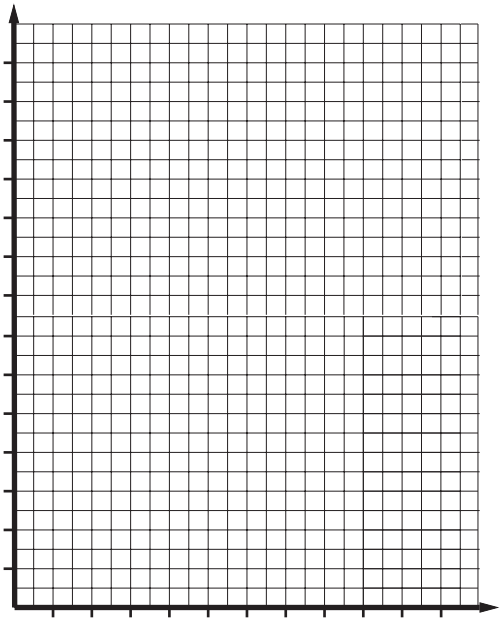
a How much more money was spent on health than on education?

b How much more money was spent on education than on crime?

c Find the total cost to the community of crime, education and health.

d What percentage of this cost is related to crime?

e Draw a picture graph to display the three costs.



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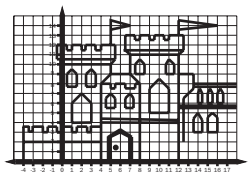
Answers

1 Robin Hood

1 784 **2** 39 **3** 135 **4** Lack of accurate records.



5 **a** 90 **b** 70 **c** 110 **d** 130

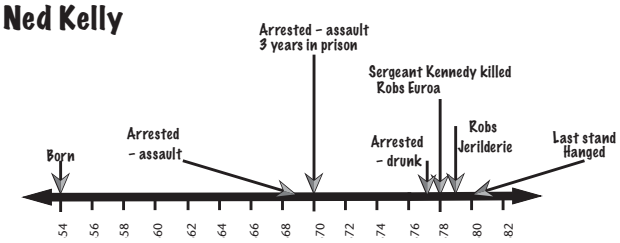
6 **a** 110 **b** 50 **c** 50 **d** 90

7 Robin Hood, he scored 400 and Little John scored 300.

8

2 Ned Kelly

1



2 16yr5m **3** 26y5m **4** 5m **5** **a** 11y11m **b** Oct 1855

c 8y4m **d** 6y2m **e** 4y5m **f** 2y3m **g** 2m

6 **a** 13y1m **b** 11y2m **c** 9y6m **d** 7y5m **e** 5y7m

f 3y5m **g** 1y4m **7** Very accurate, given that he was only 11 or 12 years old **8** I think he was born in June 1885, I think Ned should know his age better than his father does. **9** £8000 **10** $16 000 **11** $1 055 649.44

12 **a** $4 520, $4 828 **b** $307 764.04, $318 542.22

13 **a** 38 359.35 **b** $39 638.00 **c** $24 294.26

d $21 992.70 **e** $317 615.46 **f** $49 867.16

g $44 752.58 **h** $13 681.50 **i** $343 444.09 **j** $39 126.54

k $90 144.48

3 Ned Kelly rides again

1 A large dog, a small person **2** **a** 45 000 g **b** 40 000 g

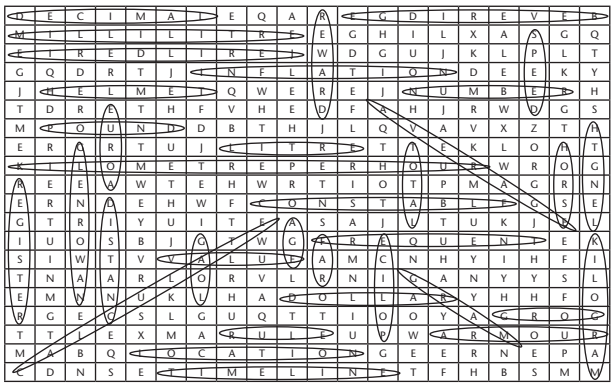
c 45 200 g **d** 46 300 g **3** **a** 3 kg **b** 48 kg **c** 4.56 kg

d 56.5 kg **4** **a** ≈40 km **b** ≈310 km **c** ≈280 km **d** ≈230 km **5** **a** ≈25 h **b** ≈7 h **c** ≈7 h **d** ≈8 h

e Ned did not stop for a rest **7** **a** Shepparton

b Jerilderie **c** Tocumwal **d** Myrtleford **e** Melbourne

8 **a** A2 **b** D4 **c** C2 **d** C4 **9**



10 Euroa, armour, reward, Jerilderie, constable, Glenrowan, horse, helmet, police, register, gaol, Beveridge, grog **11** **a** kilometreperhour **b** inflation **c** timeline

d kilogram **e** length

4 Old Melbourne Gaol

1 $40 000, $5 800 322.80 **2** **a** $2 477 557 **b** $4 459 603

c $6 193 893 **d** $2 229 801 **e** $322 082 **f** $3 964 091

g $4 211 847 **h** $23 858 875 **3** $30 million

4 **a** $48 811 **b** $25 309 **c** $32 540 **d** $27 117 **e** $18 078

f $10 847 **g** $7231 **5** Wages have increased far more than the average rate of 3.2%. People doing these jobs today would earn much more **6** $169 933

7 Gaoler and Doctor **10**

8 Superintendent and other staff

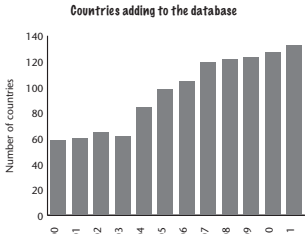
9 **a** Exercise yard

b Hospital **c** Cell block

d Governor's residence

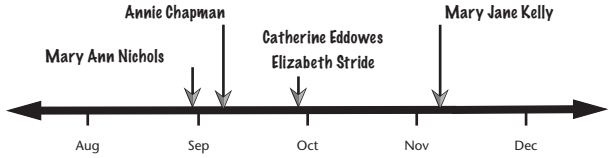
e Main entrance

11 90 **12** 66.67% **13** 70%



5 Jack the Ripper

1



2 1 3 0 5 **3** 41.40 years **4** **a** 3/20 **b** 4/15

5 **a** 15% **b** 26.67% **6** **a** 1500 h **b** 1166.67 h

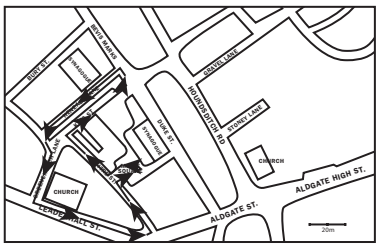
c 5166.67 h **d** 6400 h **7** **a** 1500 d **b** 1200 d

c 1950 d **d** 2190 d **8** **a** Old Montague St **b** Candle St

9 **a** Mary H6, Annie D7, Elizabeth G3, Catherine B3, Mary Jane C6 **10** London Hospital **11** ≈3.6 km

12 ≈9 km^2 **13** **a** NE **b** E **15**

c SE **d** NW **14** Travel SW along Whitechapel Rd, turn right into Commercial St, turn left into Dorset St.



16 ≈300 m **17** 3.6 m

18 133 rounds

6 Espionage

1 18/10/1909 **2** 1981 **3** 23 y **4** 18 y **5** 1935 **6** 1936

7 1942 **8** 1954 **9** 1966 **10** 75y6m **11** **a** First letter incorrect. Ciphers and codes have been around since the Romans. **b** Words are backwards. Spies have used codes since the Romans **c** Vowels are missing. The science of encoding secret messages is called cryptography.

12 **a** If you wish to be loved, love **b** He who profits by crime, commits it. **13** **a** He who does not prevent a crime when he can encourages it. **b** If you share your friend's crime, you make it your own.

7 James Bond, 007

1 22 **2** 46 **3** 1989–1985, 6y **4** *Licence to Kill* and *Golden Eye* **5** $12.25 billion **6** $557 000 000

7 *Thunderball, Goldfinger, Live and Let Die, You Only Live Twice, The Spy Who Loved Me*.

8 *Licence to Kill, A View to Kill, The Living Daylights, Dr No*. **9** **a** 1/52 **b** 1/13 **c** 1/4 **d** 1/2 **e** 2/13 **f** 2/13

10 0.05 s **11** 0.04 s **12** Yes, 0.01 s to spare. **13** No

14 29 cm **15** No, 0.02 s for bullet to reach Bond, he needs 0.04 s to reach cover. **16** 6.10 s **17** 1.25 s **18** 0.42 s **19** No **20** 3.44 m

8 Lara Croft, Tomb Raider

1 1813%, 123%, 235% **4** and **5**

2 1813% **3** 4600%

6 **a** ≈17 m **b** ≈44 m **c** ≈10 m

d ≈75 m **e** ≈48 m

7 **a** ≈12 m **b** ≈4 m **c** ≈32 m

d ≈28 m **e** ≈22 m

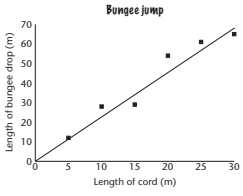
8 **a** 0.26 h **b** 0.63 h **c** 0.16 h

d 0.95 h **9** **a** 540 km **b** 756 km **c** 270 km **d** 135 km

10 **a** $159 million 138.26% **b** $61 million 64.21%

11 *Lara Croft:Tomb Raider* as % gross profit higher.

12 **a** 12 454 545 **b** 7 090 909



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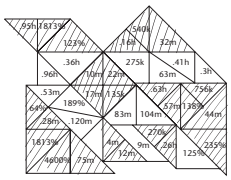
Answers (cont'd)

14 7 polygons **13**

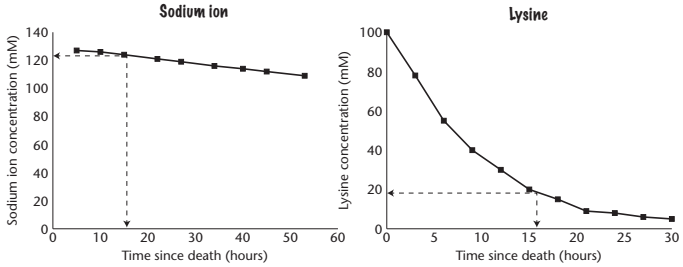


9 Time of death

1 Clothes, air temperature, wind, location (forest, desert) etc.

2 **a** 5.21 pm **b** 9.11 pm **c** 12.30 am **d** day before at 5.16 pm **e** 4.30 am **3** Willy, time of death 10.10 am.

4



5 Average time since death ≈15.5 h, time of death ≈5:30 pm **6** Tony Butcher was at the scene at the time of death of 5:30 pm **7** **a** 12 h **b** 5–7 h **c** Less than 5 h or more than 36 h **8** 31 h

10 Who dunn it

1

Real Name	Real occupation	Assumed name	Assumed role
Damon	Murderer	George	Priest
Gareth	Thief	Tyron the Terrible	Body guard
Marcus	Robber	Eric of Bondi	Businessman
Sheila	Marksman	Kelvin	Bounty hunter
Sandy Scarface	Cyber thief	Bruno	Road warden

2

First name	Surname	Crime	Current job
Alex	Jefferson	Theft	Baker
Eril	Chamming	Kidnapping	Sheep farmer
Charlie	Bartlett	Vandal	Wheat farmer
Peter	Lemming	Arson	Engineer
Andrew	Brown	Fraud	Shop owner

11 Motor vehicle crime

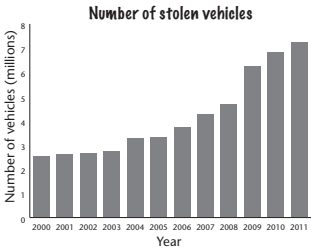
2 More cars being stolen, **1** more countries registering stolen cars. **4** 2007 **5** 2000 **6** 2004

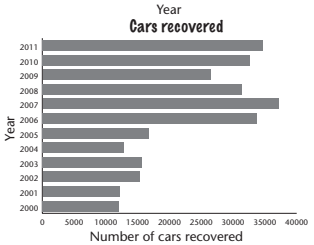
7 Nations see it as being highly useful to recover stolen vehicles, improved technology in countries.

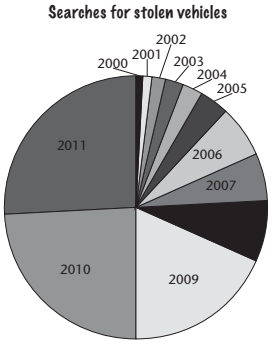
8 2004, 2007 **9** 2003

10 2001 **11** ≈122

12 **a** 4° **b** 4° **c** 6° **d** 8° **3**







e 8° **f** 13° **g** 23° **h** 21° **i** 27° **j** 66° **k** 87° **l** 92° Total searches 25 863 **13** 1000

14 82% **15** 137 500

16 1250 **17** **a** 13 200 **b** 44 000 **c** 25 300 **18** 14 300

12 Hoon drivers

1 Stopping Distances: 4.58, 14.16, 28.74, 48.32, 72.90, 102.48, 137.06, 176.64, 221.22, 270.80, 325.38.

3 18.72 m **4** 48.32 m **2**

5 29.58 m **6** 54.58 m

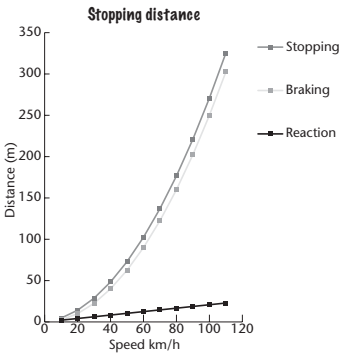
7 As speed increases the rate of stopping increases.
If you double your speed your braking distance is quadrupled, not doubled.

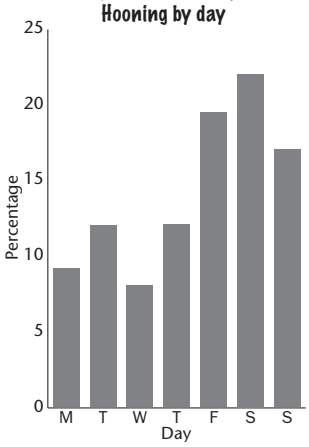
8 Talking on mobiles, kids being disruptive, eating, smoking, loud music etc.

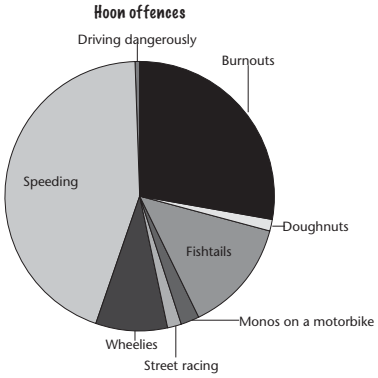
9 Good brake linings and pads, good tyres, suitable roads, dry weather conditions etc.

10 7 **11** They use 5 days per week to distort the statistics.

12 **a** 100° **b** 5° **c** 48° **d** 7° **e** 6° **f** 31° **g** 57° **h** 2° **13** 97.76% **14** 335







15 36.91% **16** **a** 266.67% **b** 233.33% **c** 70% **17** 40 **18** 89.26% **19** 228 **20** 0.22% **21** **a** 9.17% **b** 54 **c** 36 **d** 12.08% **e** 19.46% **f** 98 **g** 17.00%

13 Crash scenes

1 **a** 66.80 km/h **b** 79.04 km/h **c** 89.62 km/h **d** 94.49 km/h **e** 60.27 km/h **2** **a** 45.17 km/h **b** 56.23 km/h **c** 58.89 km/h **d** 64.83 km/h **e** 39.70 km/h **3** **a** 111.55 m **b** 192.16 m **c** 294.55 m **d** 52.72 m **e** 157.30 m **4** **a** 1003.92 m **b** 1729.41 m **c** 2650.98 m **d** 474.51 m **e** 1415.69 m **5** The car has much difficulty in stopping, there is very little friction force between the tyres and the road. **6** No, she was travelling at 53.44 km/h

7 The Commodore was only travelling at 49.22 km/h, it was not speeding, but is unroadworthy and may have stopped in time to avoid an accident if it had good tyres.

8 **a** Friday **b** Sunday **c** ≈3500 **d** Monday **e** Drivers are tired, more drivers on the road for the weekend.

9 **a** 16.00–17.59 **b** 2.00–3.59 **c** 2.00–3.59 **d** ≈3000 **e** More drivers on the roads, parents on school pick-ups, and people finish work. **10** **a** Roads and cars have been improved dramatically. **b** So that it is easy to compare the statistics from one year to the next. The growth in the number of cars on the road is considered.

14 Blood stain analysis

1 **a** 5.60 L **b** 3.62 L **c** 5.07 L **d** 4.48 L **2** **a** 43.75 kg **b** 44.50 kg **c** 70.21 kg **d** 108.40 kg **3** **a** 5.7, 3135, 2884, 219 **b** 4.5, 2475, 2277, 173 **c** 3.8, 2090, 1923, 146 **d** 4.9, 2695, 2479, 189 **e** 7.61, 4186, 3851, 293

4 **a** 31.76°, E **b** 28.27°, E **c** 25.38°, E **d** 30.00°, SE **e** 37.50°, S **f** 61.93°, NW **5** **a** 21.10°, 77.17 m, Sasha **b** 9.44°, 33.25 m, Raj **c** 18.00°, 64.98 m, Marty **d** 12.37°, 43.86 m, Joshua **e** 15.99°, 57.31 m, David

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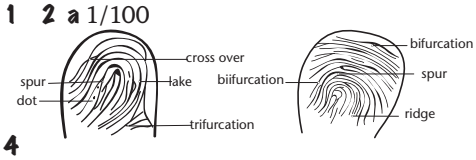
Answers (cont'd)

15 DNA

1 2.37 nanometres **2 a** 0.000 000 002 237 m **b** 2.37 × 10^{-9} m **3 a** 990 000 000 nm **b** 0.99 m **4 a** 9.9 × 10^{22} nm **b** 9.9 × 10^{13} m **c** 9.9×10^{10} km **5** 257 544 times **6** 2 505 779 m **7 a** 1.62% **b** 0.06% **8 a** 577 **b** 411y **9 a** 6 3000 000 **b** 0.64% **c** 156.96 y **10 a** 19.38% **b** 88.24% **11** Blood, hair, skin, sex cells and any other human remains **12 a** 0.42 **b** 0.40 **c** 0.14 **d** 0.13 **e** extremely small

16 Fingerprints

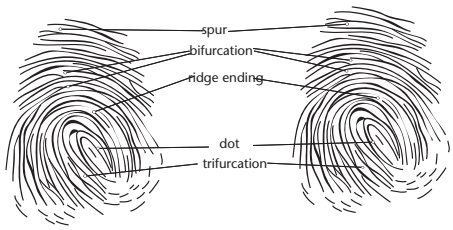
1 **2 a** 1/100



b 1/1000 **c** 1/10 000 **d** 1/10 0000 **3** 6 **4** That there is 1/10 chance linking the two prints. Not good odds to prove the two prints are identical.

5 1/100 000 000, there is a one in a hundred million chance linking the two prints.

6



Eight clear (there are others) matching minutiae have been found, giving a 1/100 000 000 match.

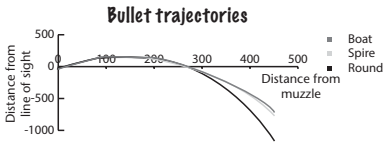
7

Bulky Bilson, ten clear matches of minutiae were found giving a 1/10 000 000 000 match.

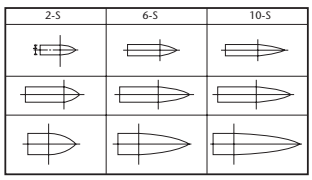
8 One in a million billion chance.

17 Ballistics

2 ≈15 m and ≈275 **1** m, the location of the target.



3 Spire point boat tail because it has the flattest trajectory and travels fastest to the target. It is best for long ranges **5** as it is also less affected by wind.

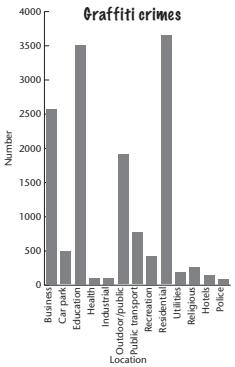


4 a 0.425 **b** 0.568 **c** 0.0159 **d** 0.0194 **e** 0.511 **f** 0.492 **g** 0.571 **h** 0.31 **i** 0.34

18 Graffiti

1 a 3640 **b** 3500 **3 a** 3.43% **b** 1.79% **c** 5.41% **d** 18.29% **4** $2.2 thousand, $1.1 thousand **5 a** 24 weeks, 12 weeks **b** 180 days, 90 days **6** $18.29 **7** 996 **8** 10 958 **10 a** 1 415 180 **b** 1 854 840 **c** 565 994 **11** 14.29% **12** 27.08% **13 a** 68.75% **b** 32.50% **14** 0.41 m^2 **15** 188 889 cans **16** $472 222.50

2

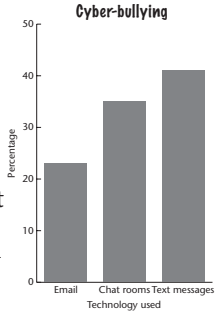


19 Cyberbullying

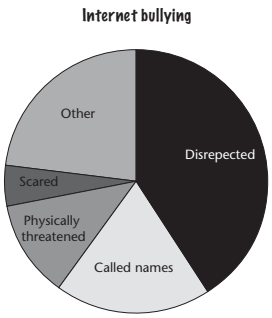
1 a 210 **b** 149 **c** 337 **d** 295 **e** 759 **2 a** 630 **b** 158 **3 a** 525 **b** 105 **4** 315 **5 a** 870 **b** 348 **6** 840 **7 a** 11, 3 **b** 9, 2 **c** 5 **d** 15, 6 **e** 15

8 Many students have been bullied by technology and many more than once, so there is a large problem.

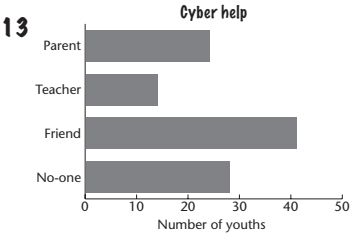
9



10 a 500 **b** 240 **11 a** 148° **b** 68° **c** 43° **d** 18° **e** 83° **12 a** 108 **b** 316 **c** 216 **14** It is important to tell someone so that together help can be obtained to cope with the stress and to stop the bullying.

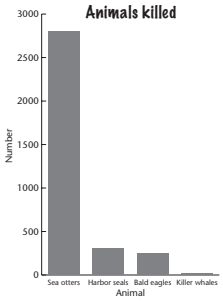


13

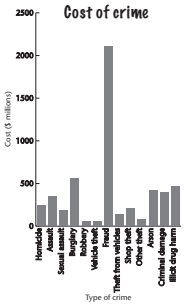


20 Environmental crime

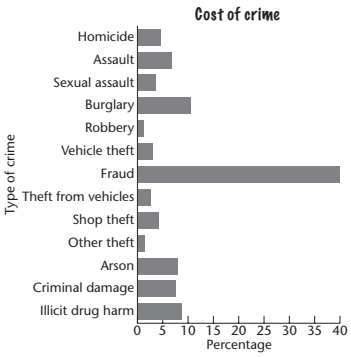
1 a 5/24 **b** 20.83% **c** 79.17% **2 a** 9660 km **b** 2093 km **c** 13/60 **d** 78.33% **3 a** 36 000 **b** 1/36 **c** 97.22% **4 a** 253 372 **b** 0.99 **c** 100/9049 **d** 0.12% **e** 98.77% **5 a** 0.83 **b** 83.33% **6** **a** $1.15 billion **b** 1/10 **7 a** 0.2 **b** 20% **8** **a** 166 666.67 ha **b** 38 461.54 ha **c** 5479.45 ha **d** 228.31 ha **9** 114.16 football fields **10** 30 million **11 a** 0.10 **b** 90.24% **12 a** 60 000 **b** 2000 **c** 720 000 **d** 13 846 **13** Total cost $3139, 0.35%, 3.82%, 7.64%, 14.91%, 73.27%.



21 Cost of crime

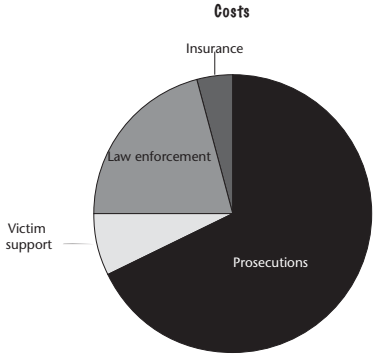


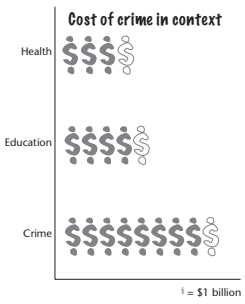
1 a 240 **b** 350 **c** 180 **d** 560 **e** 57 **f** 150 **g** 2100 **h** 130 **i** 210 **j** 74 **k** 420 **l** 390 **m** 460



2 Total $5 353 million **a** 4.5 **b** 6.59 **c** 3.36 **d** 10.44 **e** 1.06 **f** 2.80 **g** 39.77 **h** 2.43 **i** 3.98 **j** 1.38 **k** 7.77 **l** 7.34 **m** 8.57

3 a 244° **b** 27° **c** 75° **d** 14° **e** 3611





4 a Shop theft **b** Vehicle theft **c** Burglary **d** Robbery **e** Criminal damage **f** Robbery

5 a $37 billion **b** $11 billion **c** $167 billion **d** 21.56%

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